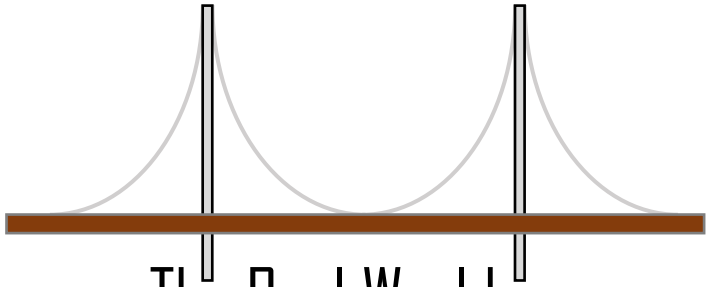
MONASH University

STATISTICS

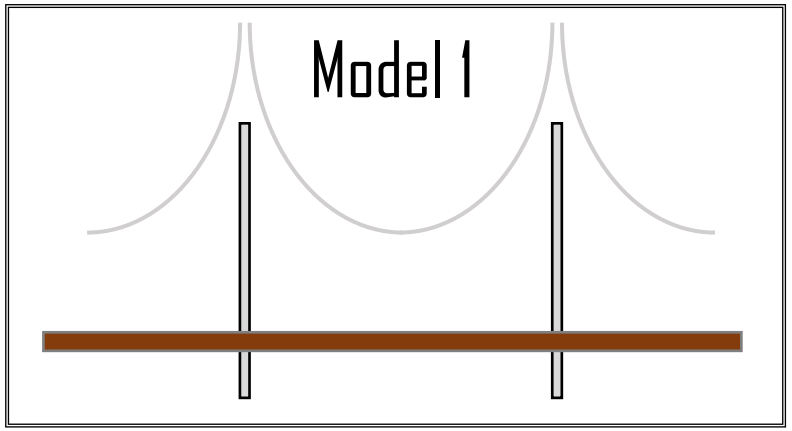
Session 7: Models & Relationships

A modelling framework

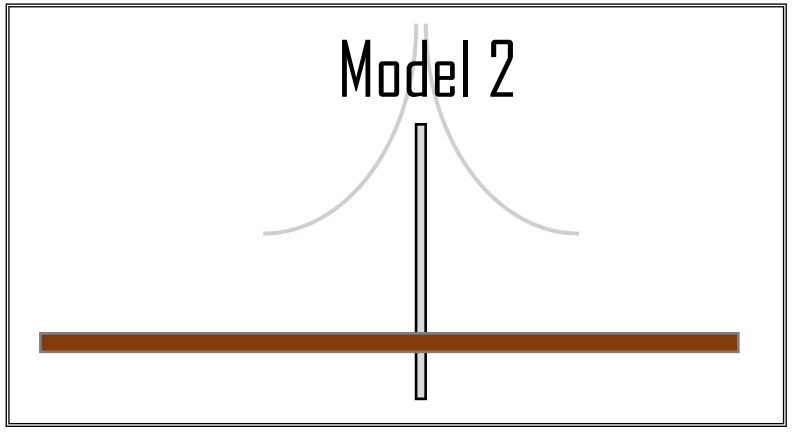


The Real World

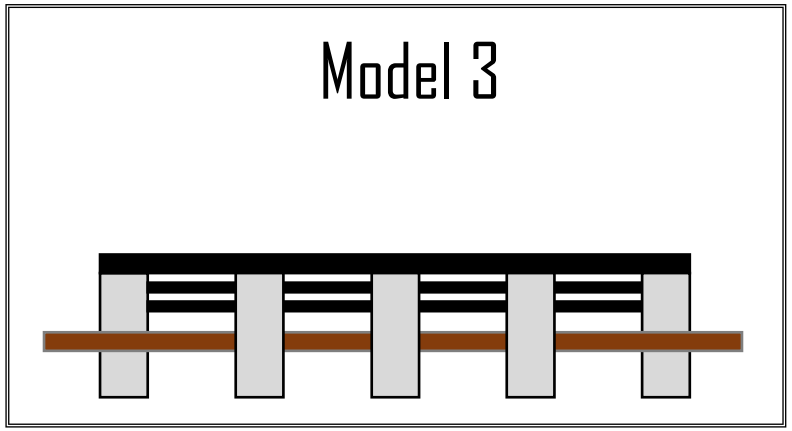
Good Fit



Moderate Fit

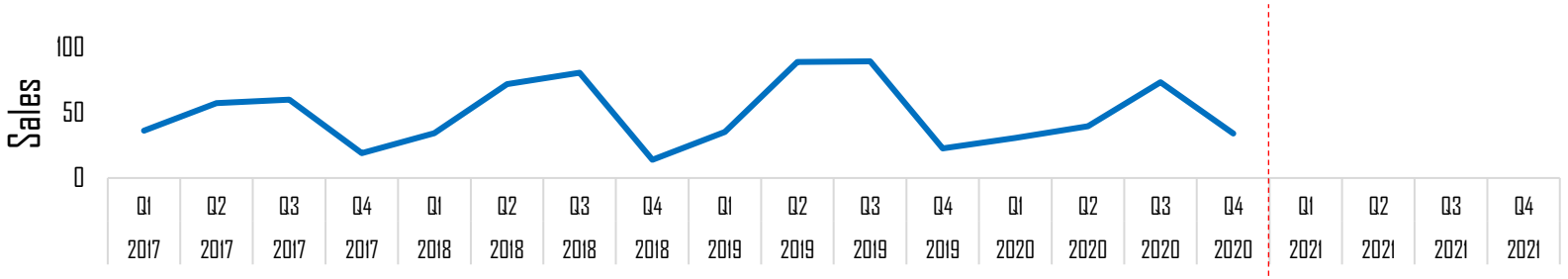


Poor Fit

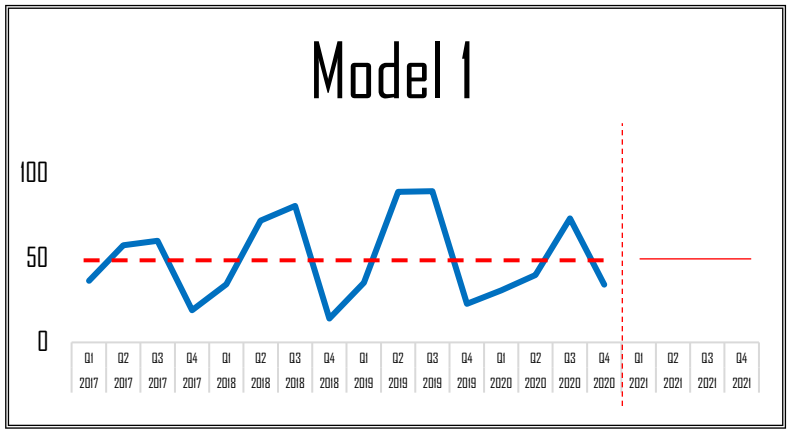


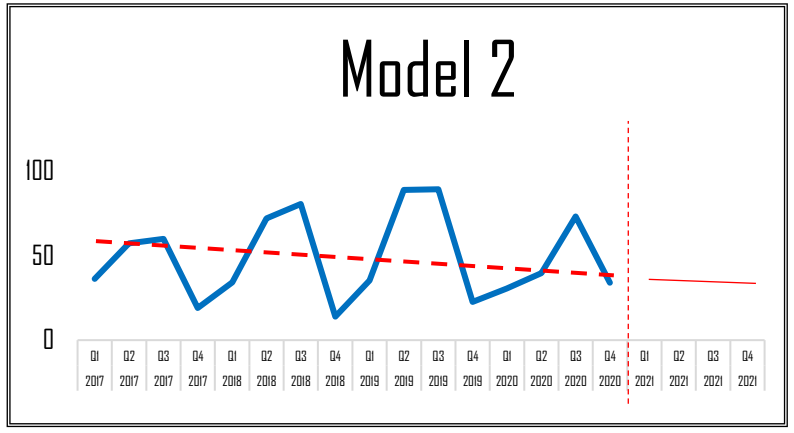
MONASH University

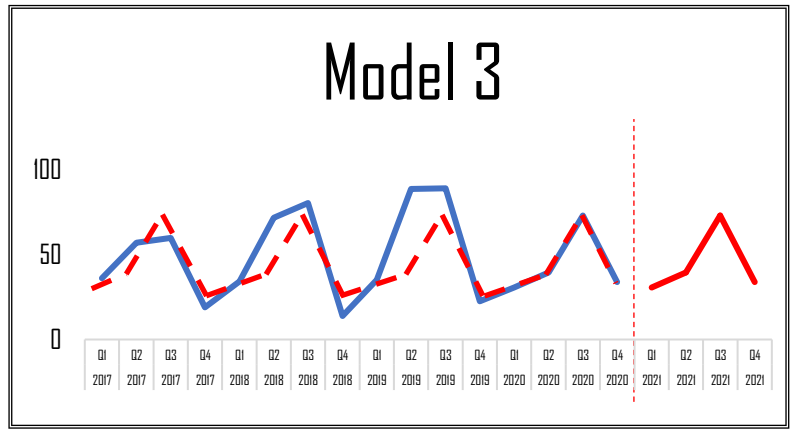
A statistical model



The Real World







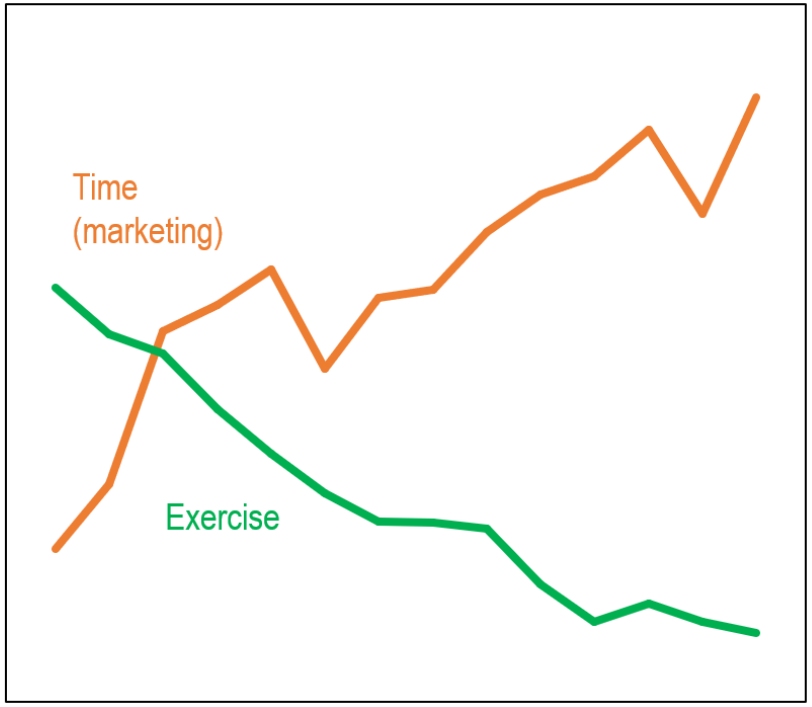
MONASH University

Covariance

Suppose a study was conducted where participants were shown varying amounts of marketing videos at the start of the week (measure as Time in hours). At the end of the week, two other variables were explored for each participant:

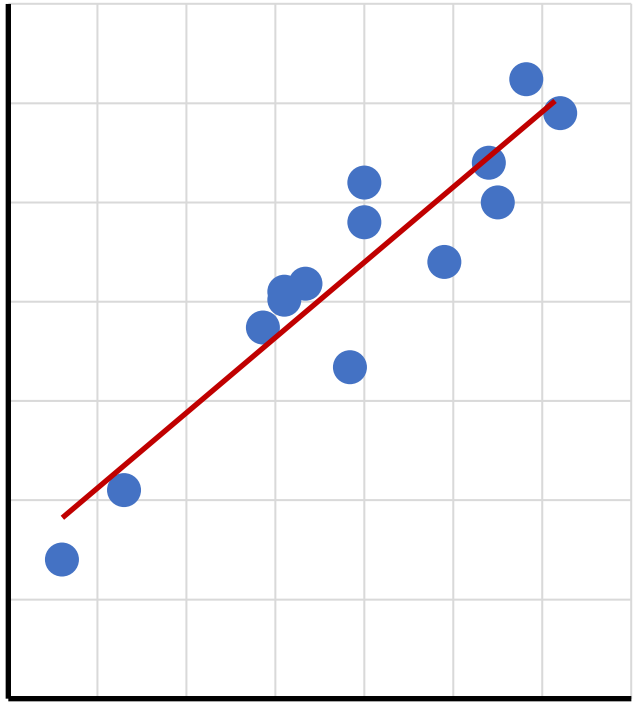
1. Amount spent ($) for online shopping that week
2. Time spent exercising (mins) that week



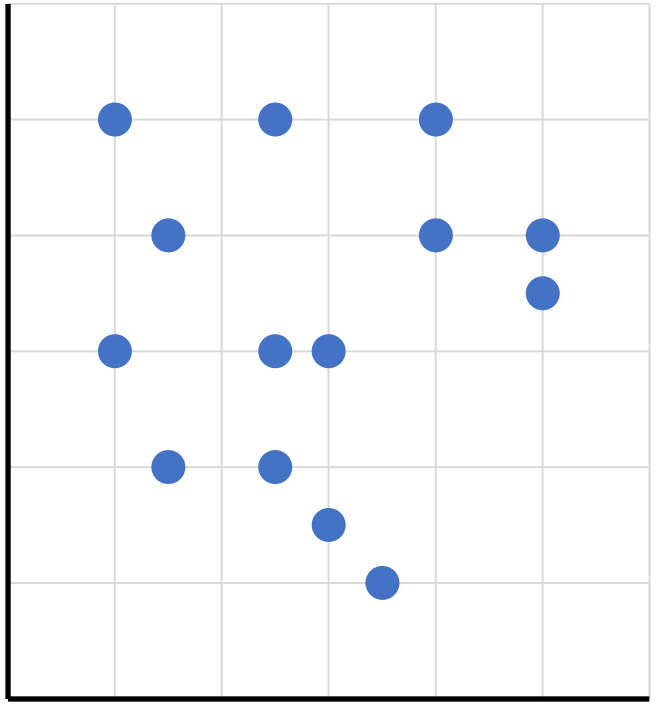


Covariance

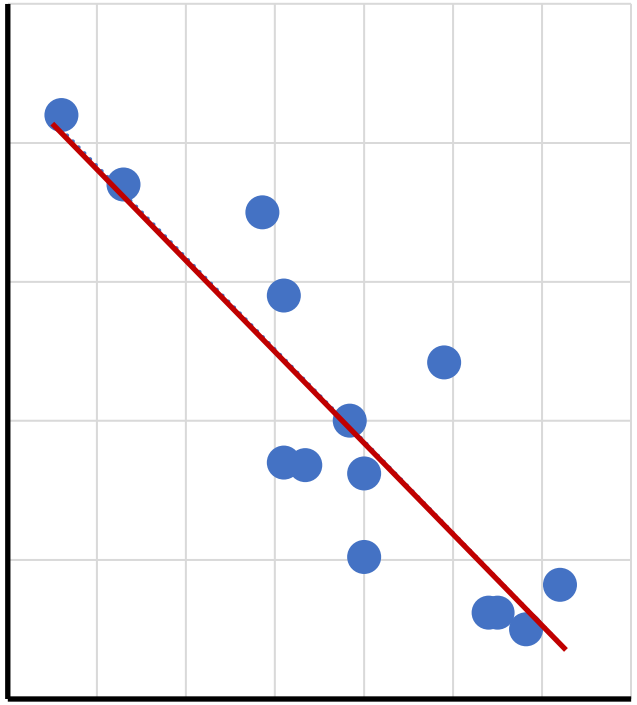
Positive



Zero



Negative



MONASH University

Covariance

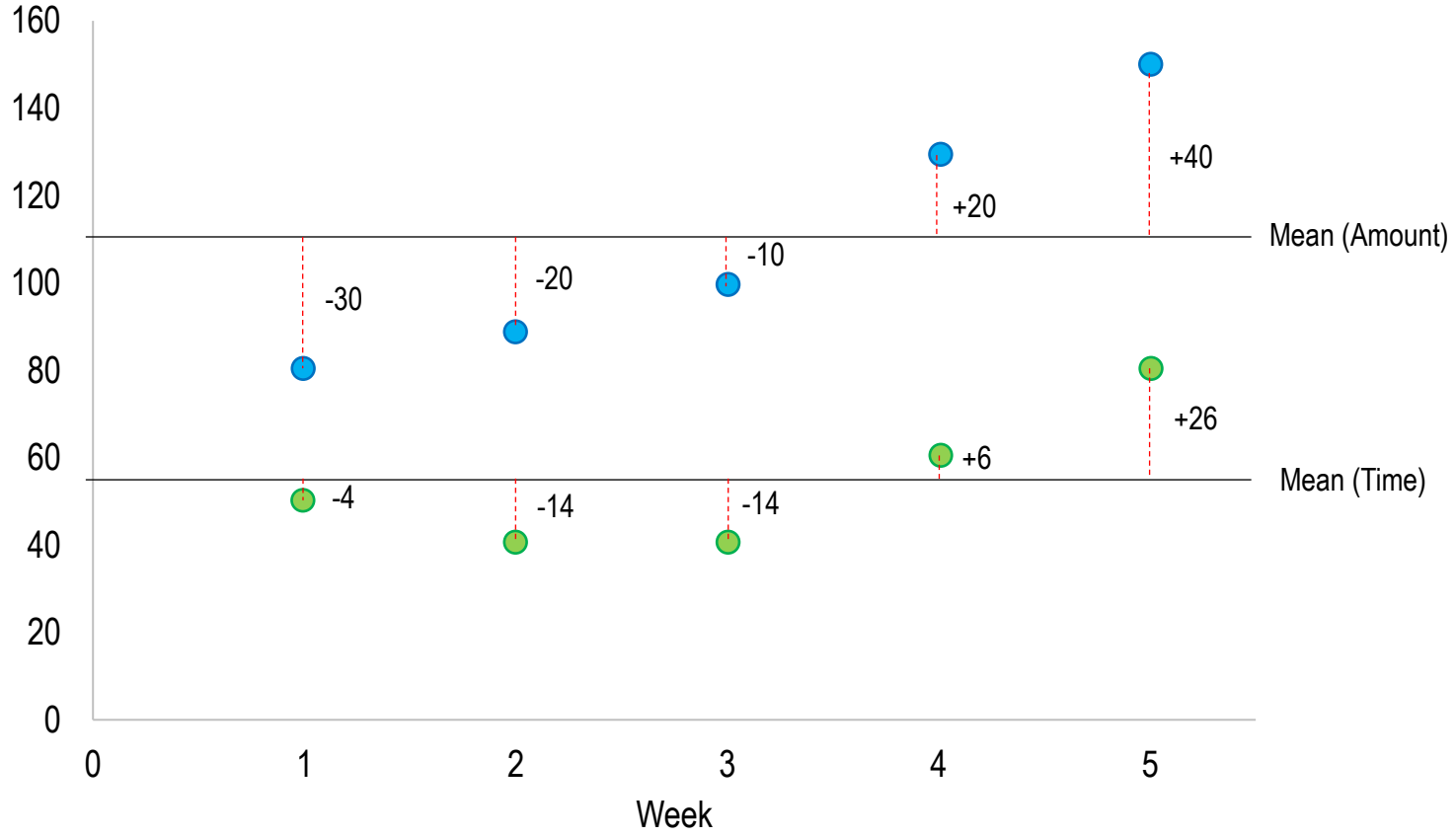
Covariance

$$\mathrm{cov}(\mathrm{x}, \mathrm{y}) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\,(y_i - \bar{y})}{N - 1}$$

MONASH University

Covariance

Time (mins)	Amount ($)
50	80
40	90
40	100
60	130
80	150
Mean: 54	Mean: 110



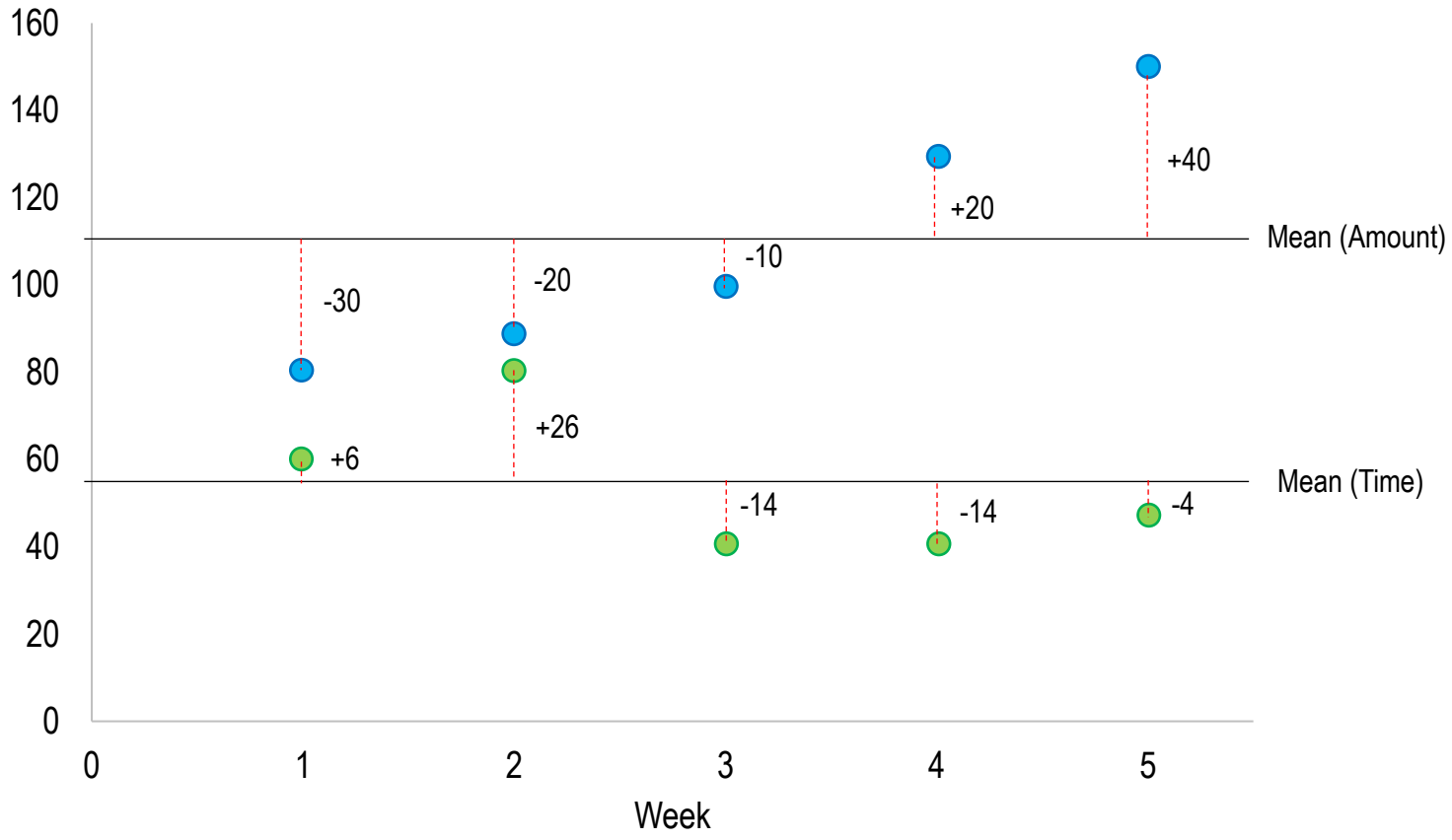
$$\textbf{Covariance}(\mathbf{x},\mathbf{y}) = \frac{\sum_{i=1}^{n}(x_i-\bar{x})\,(y_i-\bar{y})}{N-1} = \frac{(-4)(-30)+(-14)(-20)+(-14)(-10)+(6)(20)+(26)(40)}{4}$$

$$= \frac{170}{4}$$

$$= 42.5$$

Covariance

Time (mins)	Amount ($)
60	80
80	90
40	100
40	130
50	150
Mean: 54	Mean: 110

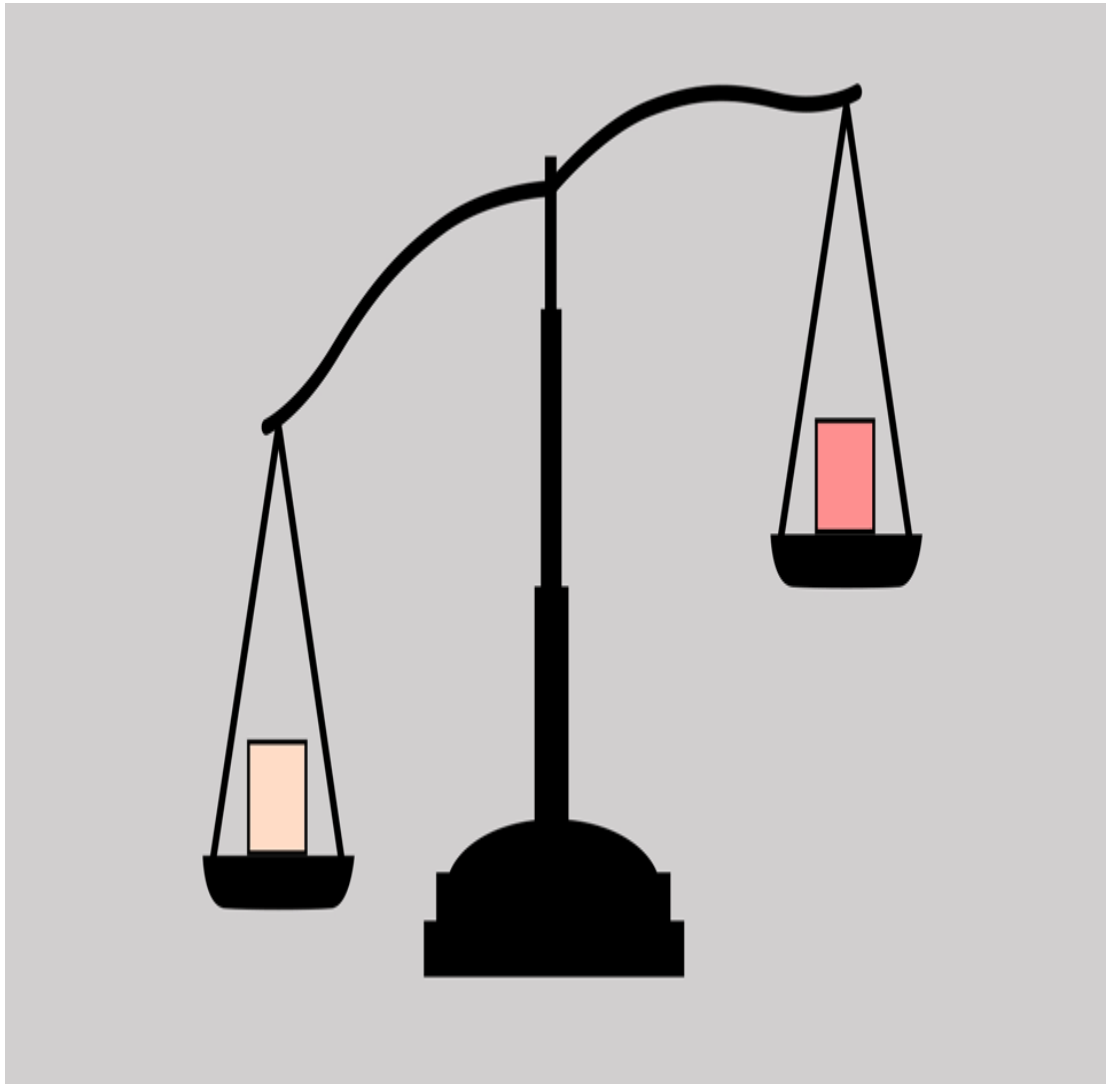


$$\textbf{Covariance}(\mathbf{x},\mathbf{y}) = \frac{\sum_{i=1}^{n}(x_i-\bar{x})\,(y_i-\bar{y})}{N-1} = \frac{(6)(-30)+(26)(-20)+(-14)(-10)+(-14)(20)+(-4)(40)}{4}$$

$$= \frac{-280}{4}$$

$$= -70$$

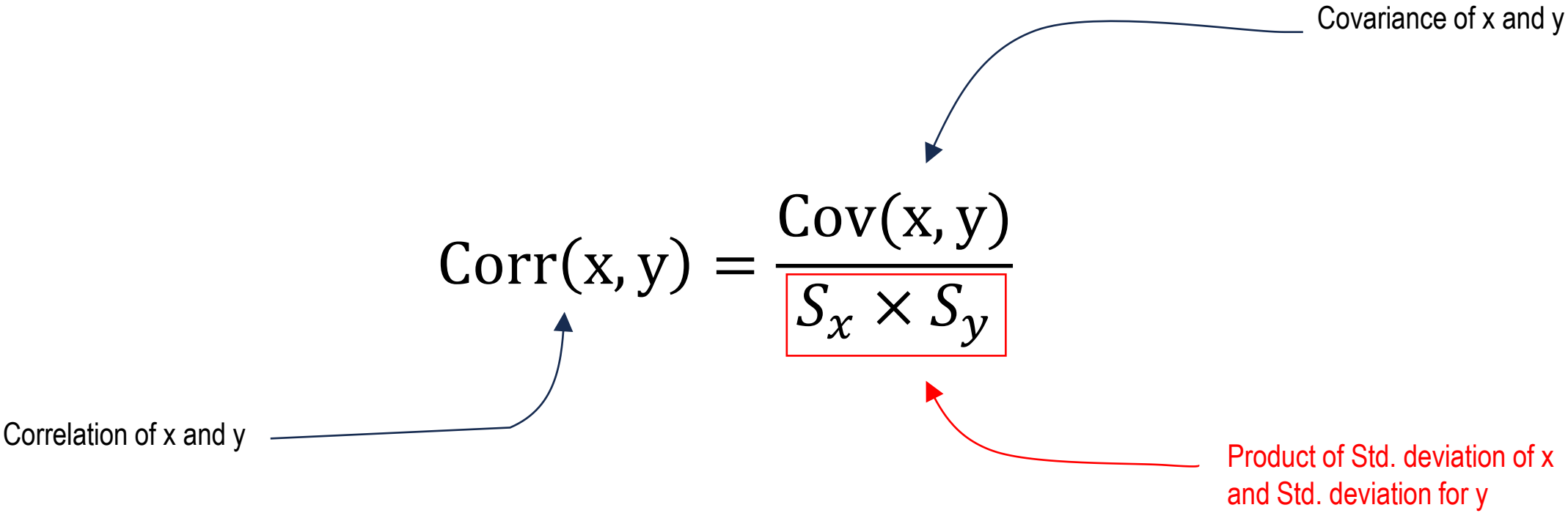
Covariance



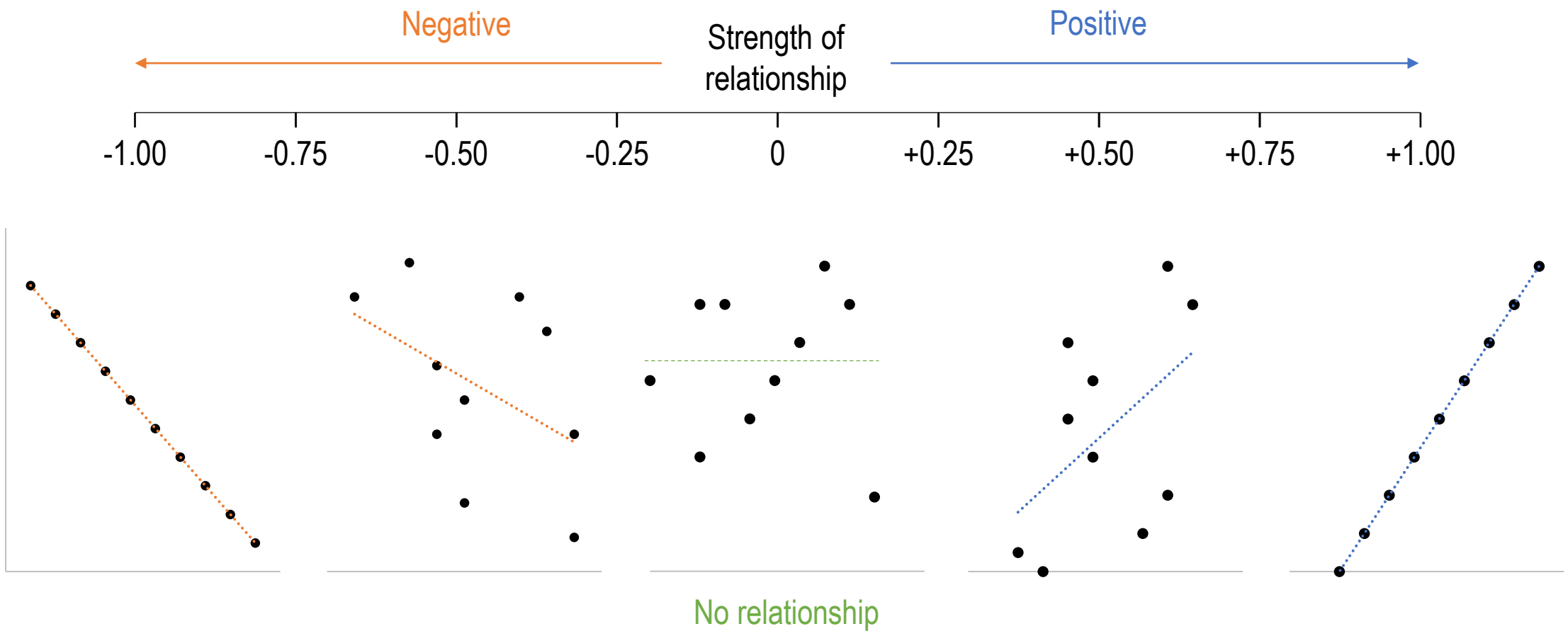
- The covariance has no built-in scale, and depends entirely on the units of your variables.
- For example, right now one of our variables is amount spent in dollars, which gave us a covariance of 42.5.
- Now, if I was to switch dollars to cents, the covariance would update as well (to 4250). It's the same data, the same relationship – however, completely different value!
- The covariance is great for identifying the direction of the relationship (positive or negative). However, the covariance is not great at telling us the strength of these relationships.

Correlation

By standardising the covariance, we obtain the correlation coefficient – a metric that provides both direction and strength between two variables. Except, unlike covariance, the interpretation of the strength is meaningful.

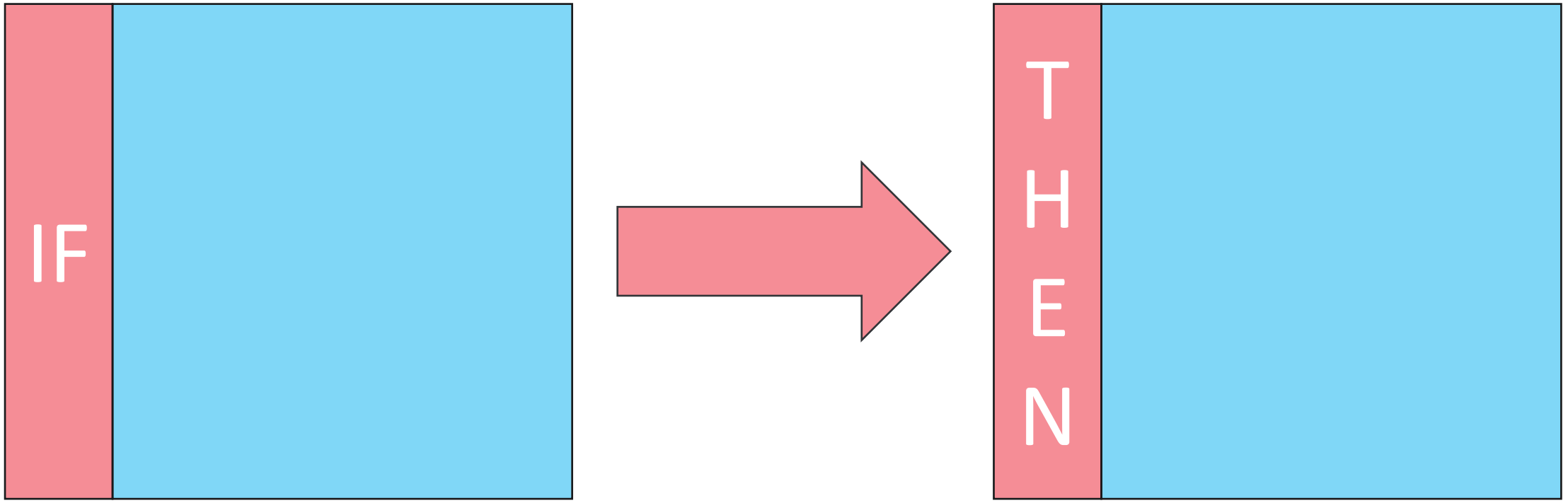


Correlation



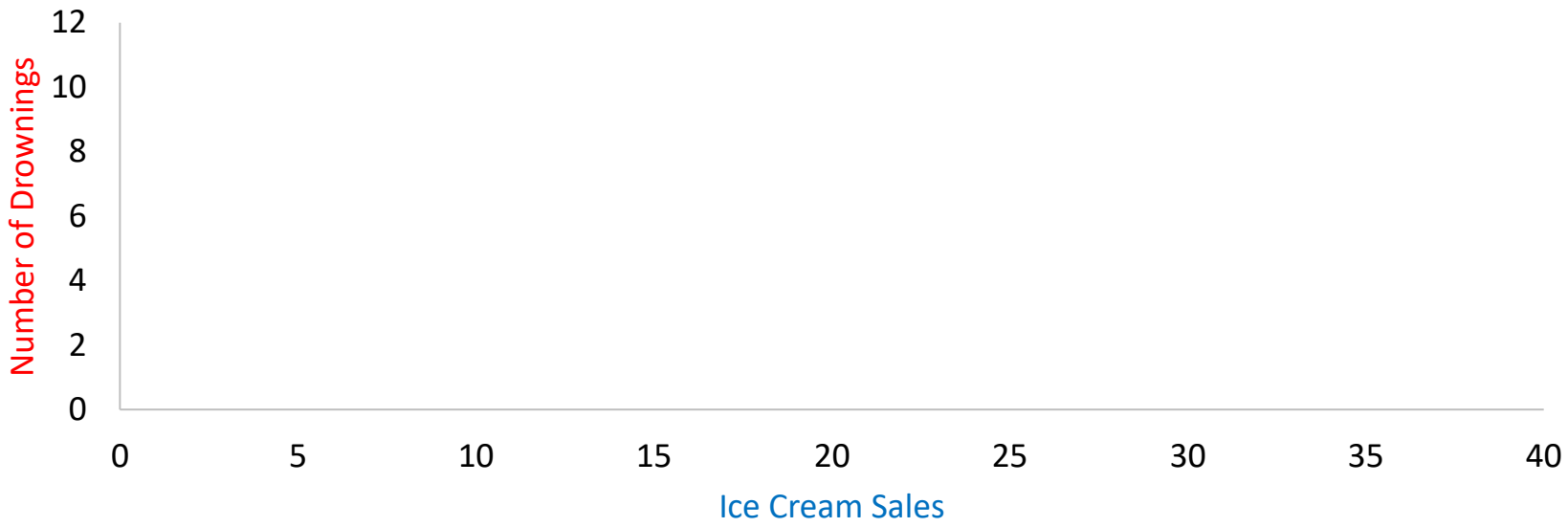
MONASH University

Correlation



Correlation is not Causation

Sales	Drownings	Sales	Drownings
12	1	22	5
14	2	22	6
13	2	26	7
15	1	29	5
15	3	31	6
18	4	30	7
19	3	32	8
22	4	34	9
23	5	35	10



MONASH University

Correlation is not Causation

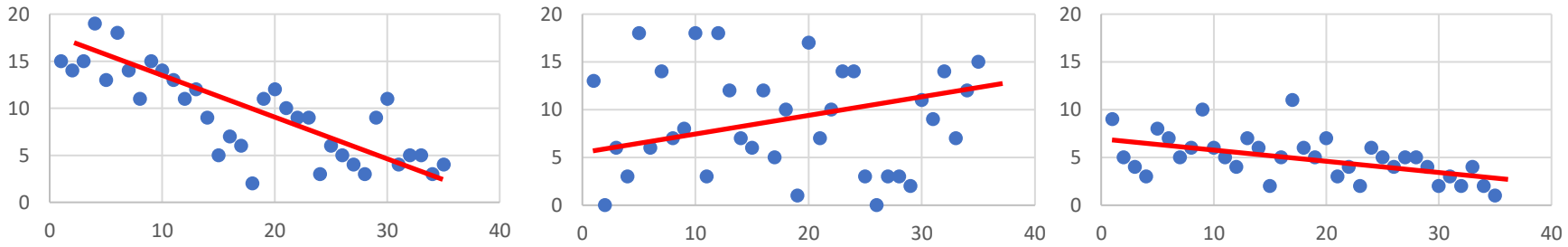






Linear Regression

- **Regression:** a procedure for finding a straight line that comes as close as possible to all the data points in a scatterplot



- The **regression equation** is a formula that describes the regression line and has the basic formula:

$$y = b_0 + b_1 x_1 + error$$

slope

Intercept / constant

- Note: When we are only dealing with two variables, this is called **simple linear regression**

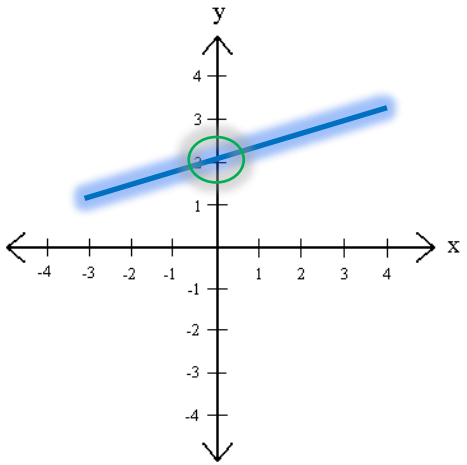
Linear Regression

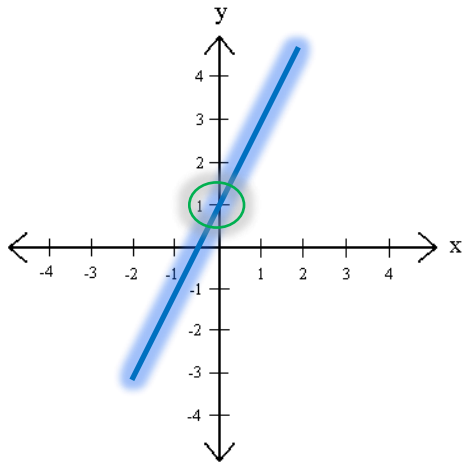
$$y = b_0 + b_1 x_1$$

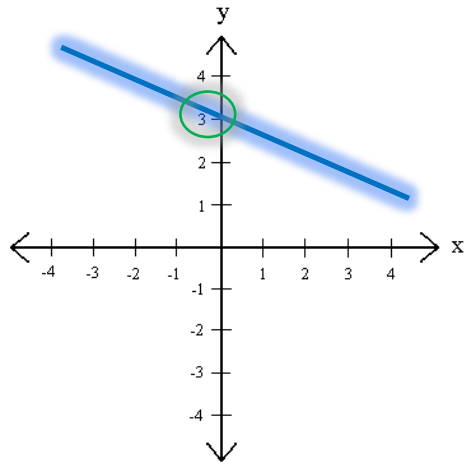
slope

Intercept / constant

The intercept (b_0) tells us at what particular point does the regression line cross the *y*-axis when *x* = 0. Because of this, it is sometimes referred to as the y-intercept.







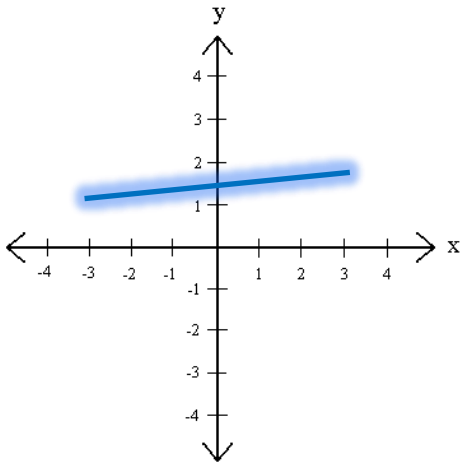
Linear Regression

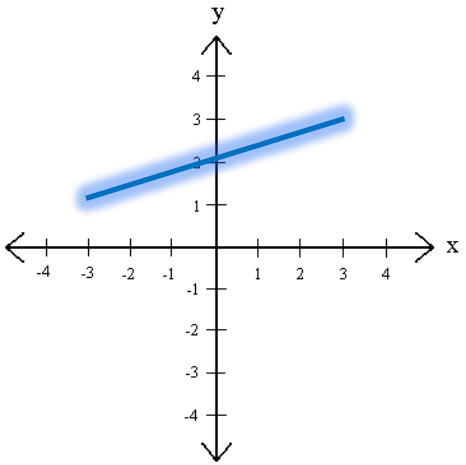
$$y = b_0 + b_1 x_1$$

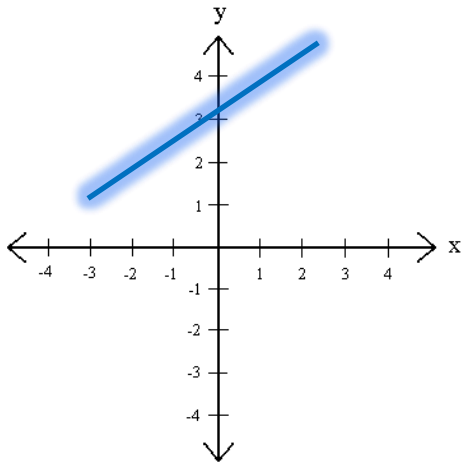
slope

Intercept / constant

The slope (b_1) tells us how much of an increase there is for *y* when the *x* increases by one unit. It is also an indication of how steep the line is.







Ordinary Least Squares (OLS)

N Friends Depression

x	y	xy	x^2	y^2
8	2	16	64	4
1	9	9	1	81
5	4	20	25	16
10	3	30	100	9
5	5	25	25	25
8	4	32	64	16
6	3	18	36	9
5	5	25	25	25
6	4	24	36	16
10	1	10	100	1
64	40	209	476	202
$\sum x$	$\sum y$	$\sum xy$	$\sum x^2$	$\sum y^2$

$$b_0 = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$= \frac{(40)(476) - (64)(209)}{10(476) - (64)^2}$$

$$= 8.53$$

$$b_1 = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

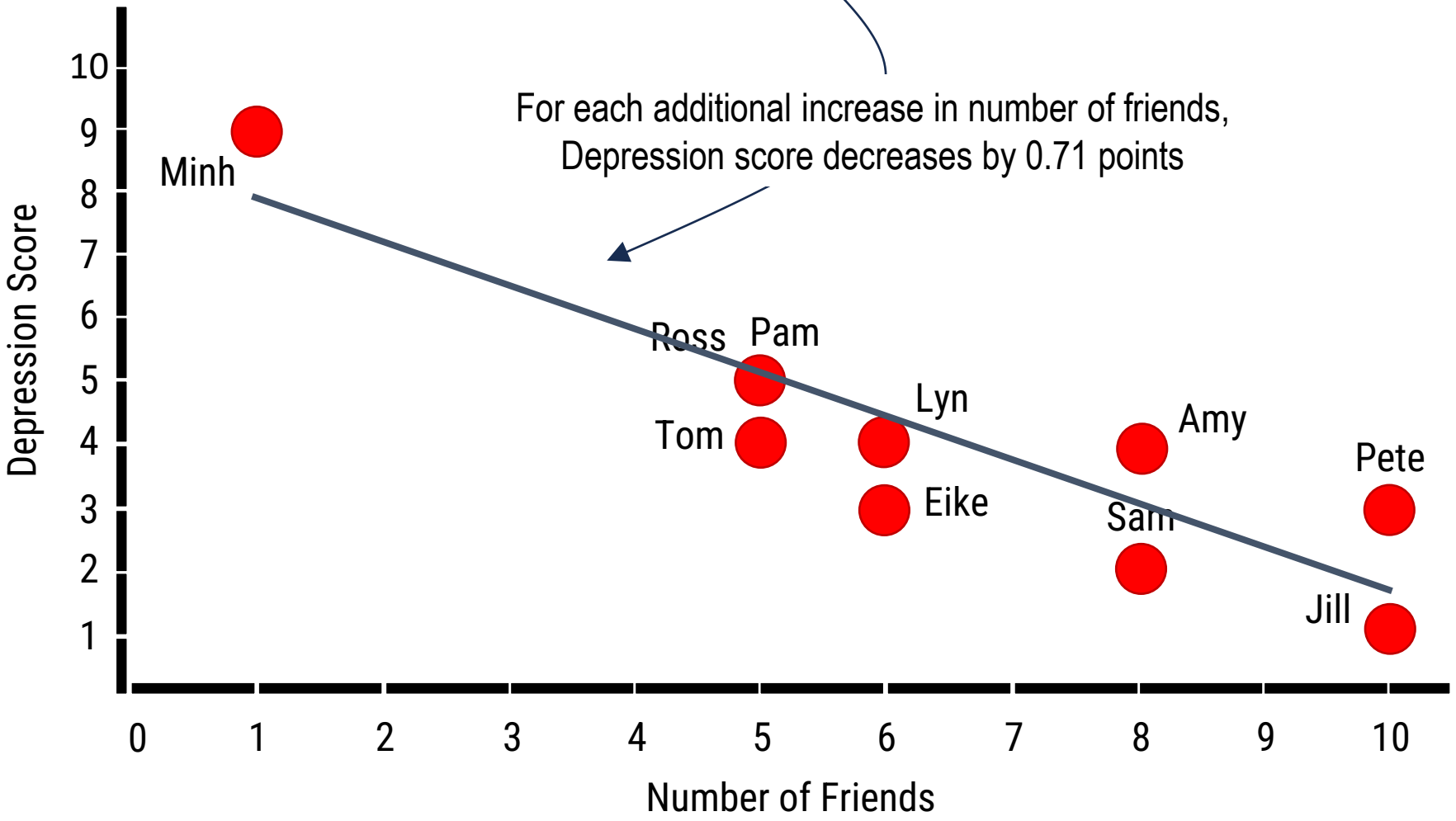
$$= \frac{10(209) - (64)(40)}{10(476) - (64)^2}$$

$$= -0.71$$

$$\text{Depression} = 8.53 - 0.71(\text{N Friends}) + \text{Error}$$

Ordinary Least Squares (OLS)

$$\text{Depression} = 8.53 - 0.71(\text{N Friends}) + \text{Error}$$



Multiple Regression

Simple Linear Regression

$$y = b_0 + b_1 x_1 + error$$

Multiple Linear Regression

$$y = b_0 + b_1 x_1 + b_2 x_2 + \cdots b_k x_k + error$$

MONASH University

Multiple Regression

Suppose we determined a multiple regression model as:

$$\text{Income} = -117119 + (4540 \times \text{Education}) + (3368 \times \text{Age}) + \text{error}$$

For each additional year in Education, income increases by $4540, when we hold other variables constant.

For each additional year in Age, income increases by $3368, when we hold other variables constant.

Multiple Regression

$$\text{Income} = -117119 + (4540 \times \text{Education}) + (3368 \times \text{Age}) + \text{error}$$

Person A



Years Edu: 12
Age: 40

$$\begin{aligned}\text{Income} &= -117119 + (4540 \times \text{Education}) + (3368 \times \text{Age}) + \text{error} \\ &= -117119 + (4540 \times 12) + (3368 \times 40) + \text{error} \\ &= -117119 + (54480) + (134720) + \text{error} \\ &= \$72{,}081\end{aligned}$$

Person B



Years Edu: 13
Age: 40

$$\begin{aligned}\text{Income} &= -117119 + (4540 \times \text{Education}) + (3368 \times \text{Age}) + \text{error} \\ &= -117119 + (4540 \times 13) + (3368 \times 40) + \text{error} \\ &= -117119 + (59020) + (134720) + \text{error} \\ &= \$76{,}621\end{aligned}$$

MONASH University

Multiple Regression

$$\text{Income} = -117119 + (4540 \times \text{Education}) + (3368 \times \text{Age}) + \text{error}$$

Person A



Years Edu: 12

Age: 40

$$\text{Income} = -117119 + (4540 \times \text{Education}) + (3368 \times \text{Age}) + \text{error}$$

$$\text{Income} = -117119 + (4540 \times 12) + (3368 \times 40) + \text{error}$$

$$\text{Income} = -117119 + (54480) + (134720) + \text{error}$$

$$= \$72{,}081$$

Person B



Years Edu: 13

Age: 40

$$\text{Income} = -117119 + (4540 \times \text{Education}) + (3368 \times \text{Age}) + \text{error}$$

$$\text{Income} = -117119 + (4540 \times 13) + (3368 \times 40) + \text{error}$$

$$\text{Income} = -117119 + (59020) + (134720) + \text{error}$$

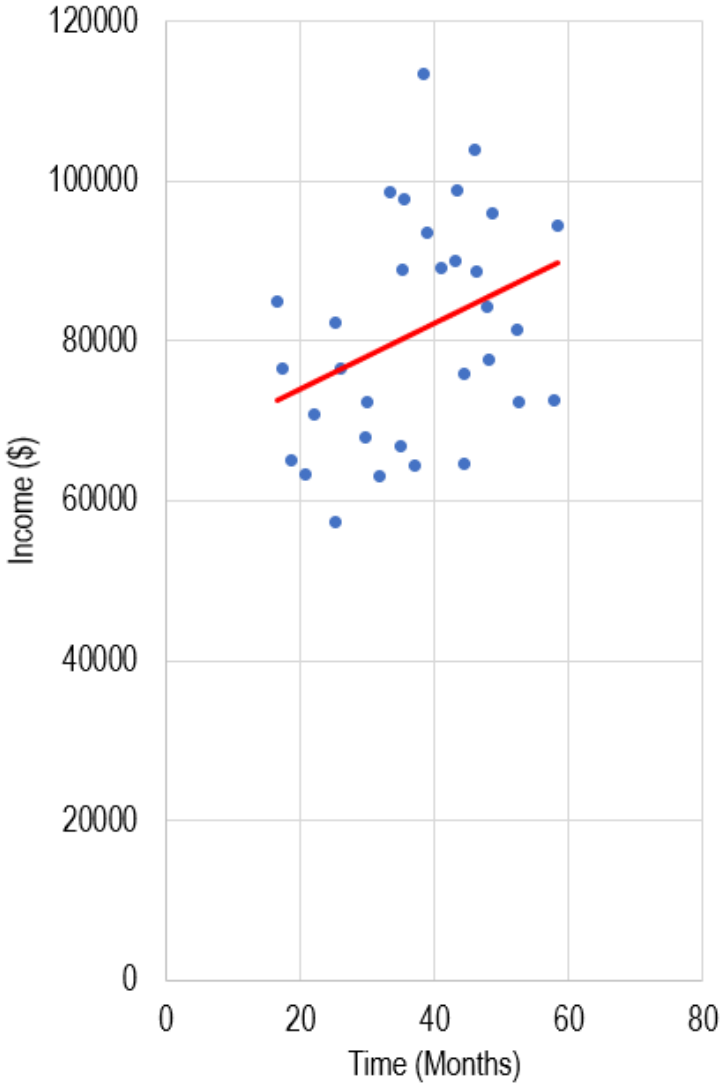
$$= \$76{,}621$$

MONASH University

R^2 and Goodness of Fit

- R^2 (also called coefficient of determination) is a statistical measure of how well the regression model explains the variability of the response variable.
  - In simple linear regression, R^2 is the square of the correlation coefficient between the observed and predicted values.
  - In multiple regression, it is the proportion of variance in the response variable explained collectively by all predictors.

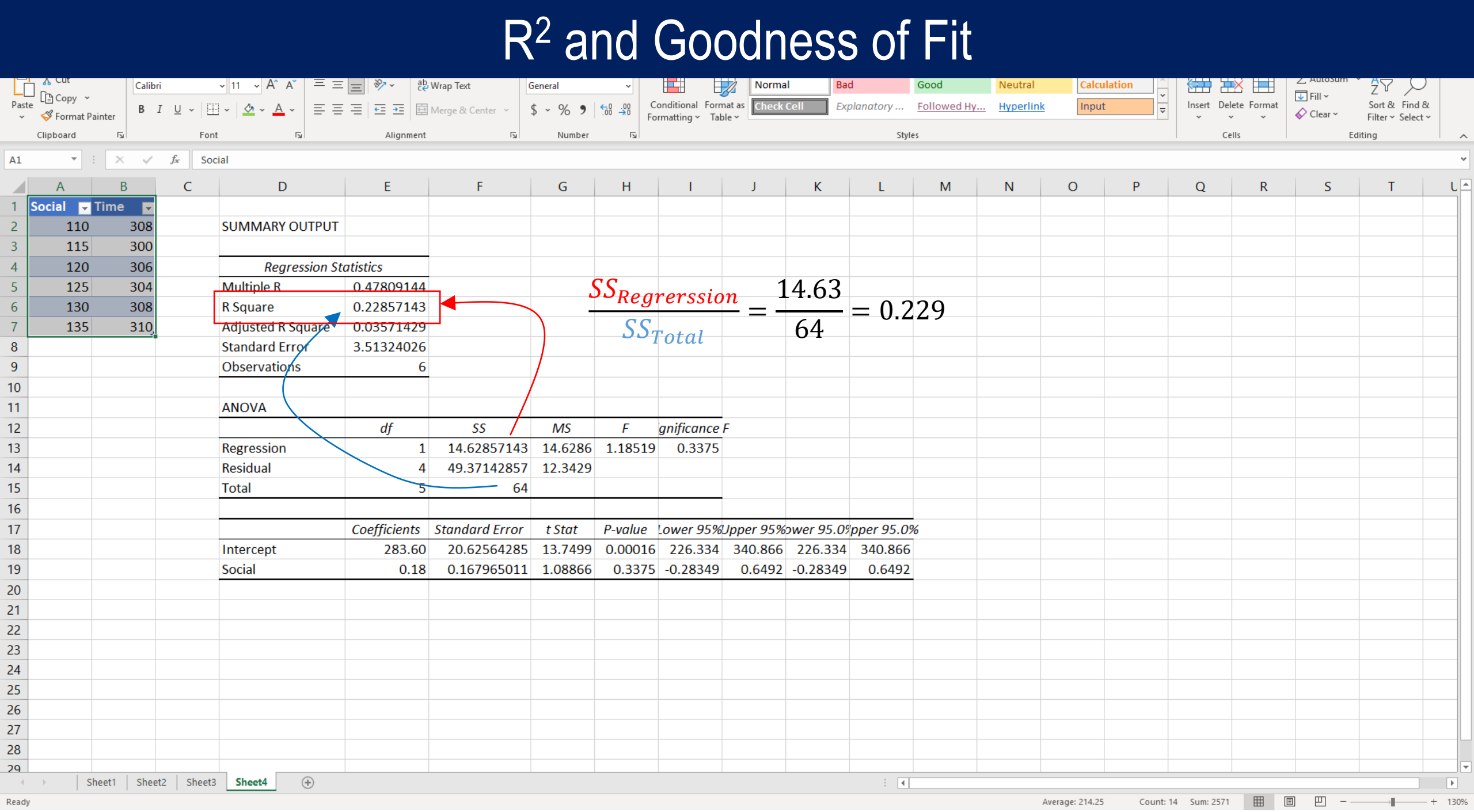
- R^2 is always between 0 and 1:
  - In general*, the higher the R^2, the better the model fits your data.



MONASH University

* Need to be mindful of overfitting models

R^2 and Goodness of Fit



Social	Time
110	308
115	300
120	306
125	304
130	308
135	310

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.47809144
R Square	0.22857143
Adjusted R Square	0.03571429
Standard Error	3.51324026
Observations	6

ANOVA

	df	*SS*	*MS*	*F*	*gnificance F*
Regression	1	14.62857143	14.6286	1.18519	0.3375
Residual	4	49.37142857	12.3429		
Total	5	64			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	283.60	20.62564285	13.7499	0.00016	226.334	340.866	226.334	340.866
Social	0.18	0.167965011	1.08866	0.3375	-0.28349	0.6492	-0.28349	0.6492

$$\frac{SS_{Regrerssion}}{SS_{Total}} = \frac{14.63}{64} = 0.229$$

R^2 and Goodness of Fit

Social	Time
110	308
115	300
120	306
125	304
130	308
135	310

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.47809144
R Square	0.22857143
Adjusted R Square	0.03571429
Standard Error	3.51324026
Observations	6

Note: R^2 is suitable for you are dealing with simple linear regression

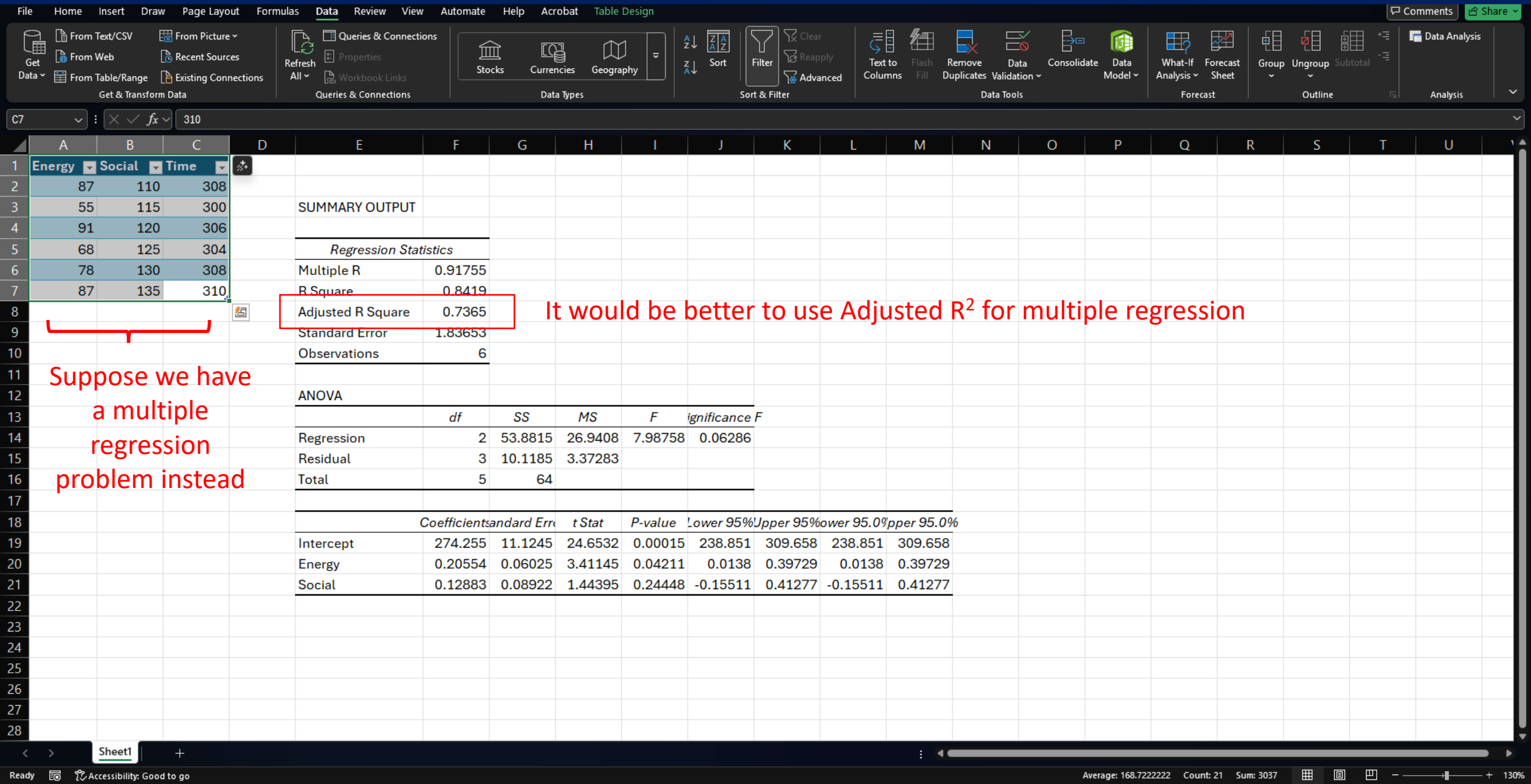
ANOVA

	df	*SS*	*MS*	*F*	*gnificance F*
Regression	1	14.62857143	14.6286	1.18519	0.3375
Residual	4	49.37142857	12.3429		
Total	5	64			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	283.60	20.62564285	13.7499	0.00016	226.334	340.866	226.334	340.866
Social	0.18	0.167965011	1.08866	0.3375	-0.28349	0.6492	-0.28349	0.6492

Sheet1 Sheet2 Sheet3 Sheet4
Ready
Average: 214.25 Count: 14 Sum: 2571 130%

Adjusted R^2 and Multiple Regression



Energy	Social	Time
87	110	308
55	115	300
91	120	306
68	125	304
78	130	308
87	135	310

Suppose we have a multiple regression problem instead

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.91755
R Square	0.8419
Adjusted R Square	0.7365
Standard Error	1.83653
Observations	6

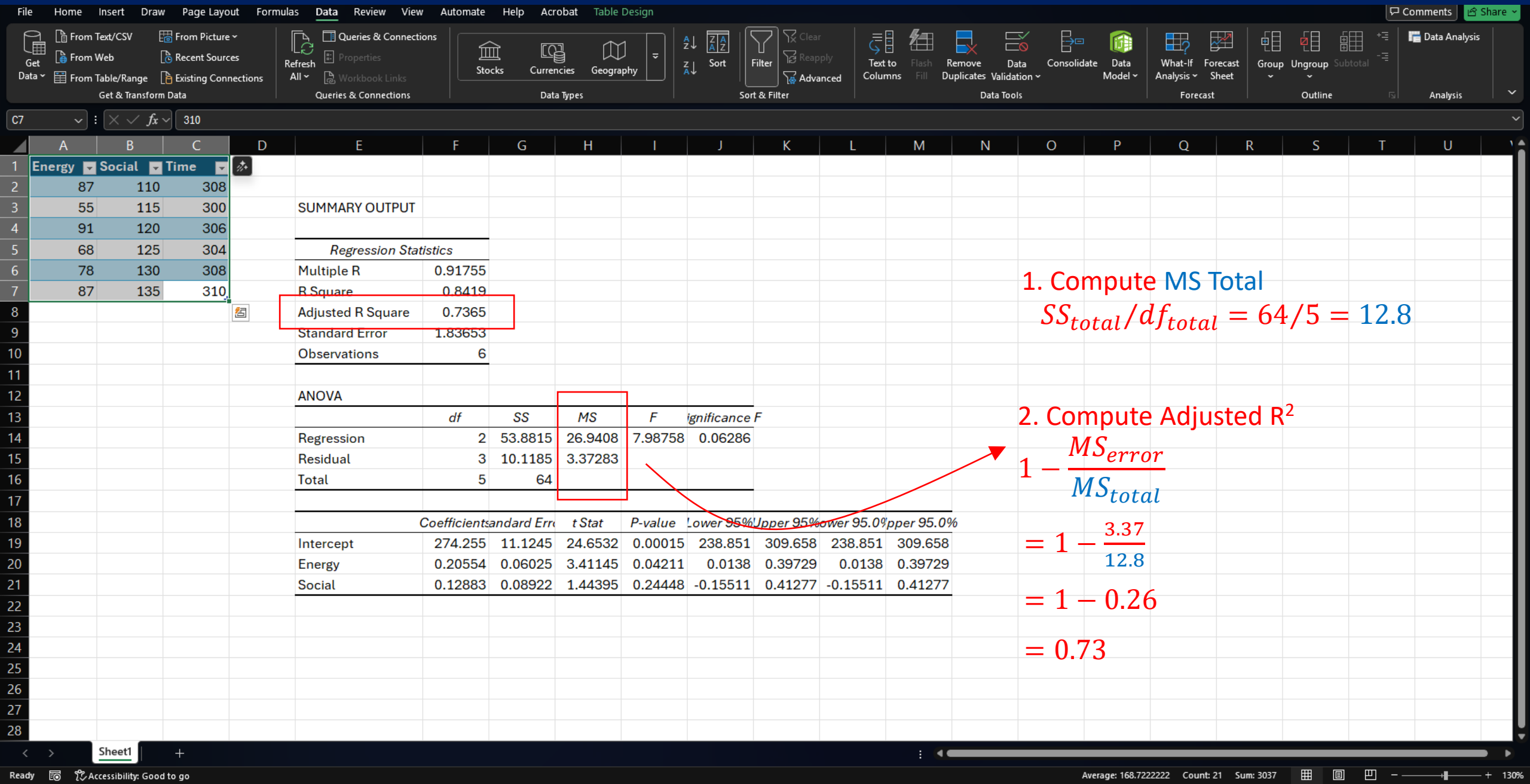
ANOVA

	df	*SS*	*MS*	*F*	*ignificance F*
Regression	2	53.8815	26.9408	7.98758	0.06286
Residual	3	10.1185	3.37283		
Total	5	64			

	Coefficients	*tandard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	274.255	11.1245	24.6532	0.00015	238.851	309.658	238.851	309.658
Energy	0.20554	0.06025	3.41145	0.04211	0.0138	0.39729	0.0138	0.39729
Social	0.12883	0.08922	1.44395	0.24448	-0.15511	0.41277	-0.15511	0.41277

It would be better to use Adjusted R^2 for multiple regression

Adjusted R^2 and Multiple Regression



Energy	Social	Time
87	110	308
55	115	300
91	120	306
68	125	304
78	130	308
87	135	310

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.91755
R Square	0.8419
Adjusted R Square	0.7365
Standard Error	1.83653
Observations	6

ANOVA

	df	*SS*	*MS*	*F*	*ignificance F*
Regression	2	53.8815	26.9408	7.98758	0.06286
Residual	3	10.1185	3.37283		
Total	5	64			

	Coefficients	*tandard Err*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	274.255	11.1245	24.6532	0.00015	238.851	309.658	238.851	309.658
Energy	0.20554	0.06025	3.41145	0.04211	0.0138	0.39729	0.0138	0.39729
Social	0.12883	0.08922	1.44395	0.24448	-0.15511	0.41277	-0.15511	0.41277

1. Compute MS Total

$$SS_{total}/df_{total} = 64/5 = 12.8$$

2. Compute Adjusted R^2

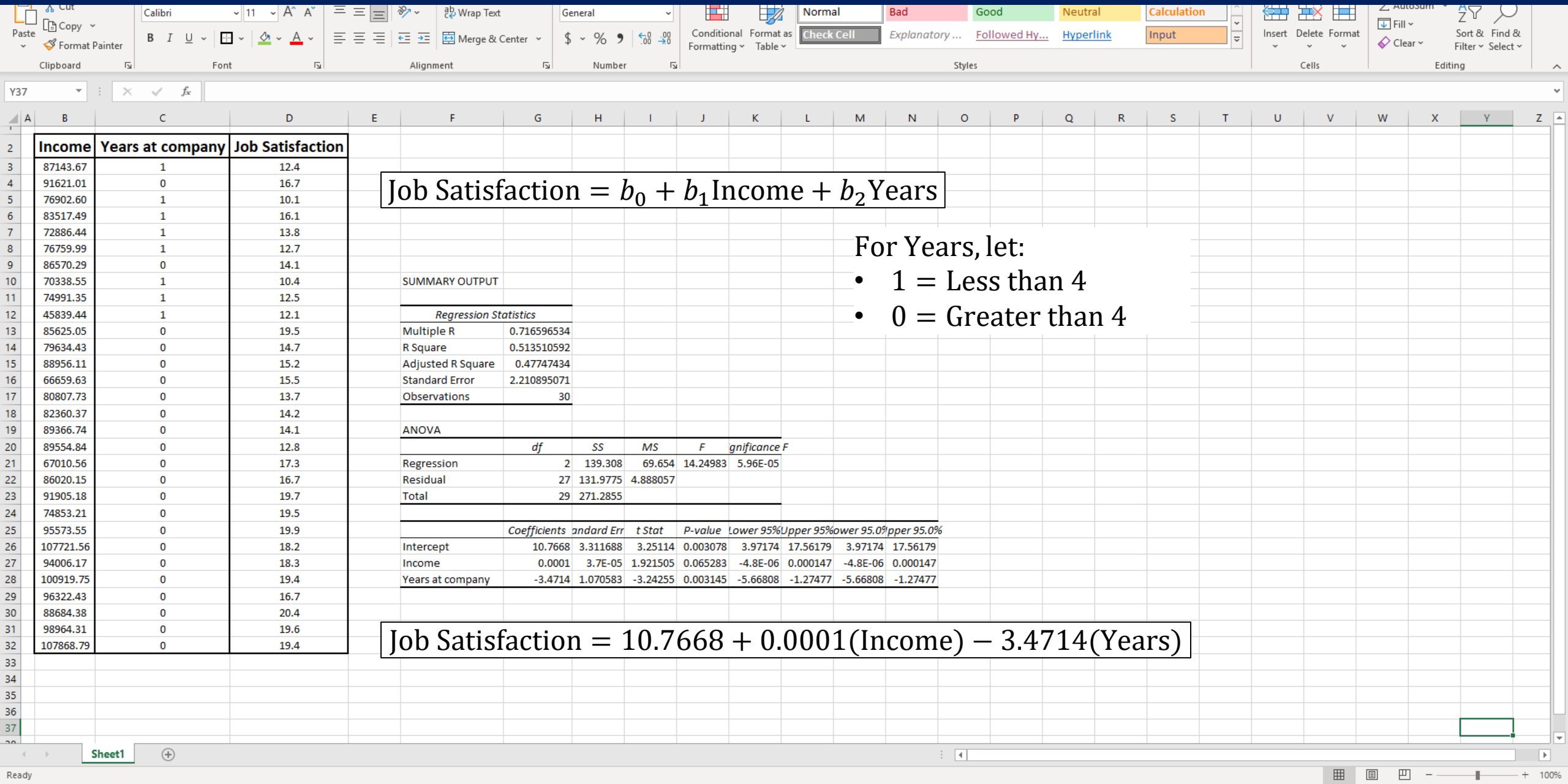
$$1 - \frac{MS_{error}}{MS_{total}}$$

$$= 1 - \frac{3.37}{12.8}$$

$$= 1 - 0.26$$

$$= 0.73$$

Multiple Regression with a binary predictor



Income	Years at company	Job Satisfaction
87143.67	1	12.4
91621.01	0	16.7
76902.60	1	10.1
83517.49	1	16.1
72886.44	1	13.8
76759.99	1	12.7
86570.29	0	14.1
70338.55	1	10.4
74991.35	1	12.5
45839.44	1	12.1
85625.05	0	19.5
79634.43	0	14.7
88956.11	0	15.2
66659.63	0	15.5
80807.73	0	13.7
82360.37	0	14.2
89366.74	0	14.1
89554.84	0	12.8
67010.56	0	17.3
86020.15	0	16.7
91905.18	0	19.7
74853.21	0	19.5
95573.55	0	19.9
107721.56	0	18.2
94006.17	0	18.3
100919.75	0	19.4
96322.43	0	16.7
88684.38	0	20.4
98964.31	0	19.6
107868.79	0	19.4

$$\text{Job Satisfaction} = b_0 + b_1\text{Income} + b_2\text{Years}$$

For Years, let:

- 1 = Less than 4
- 0 = Greater than 4

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.716596534
R Square	0.513510592
Adjusted R Square	0.47747434
Standard Error	2.210895071
Observations	30

ANOVA

	df	*SS*	*MS*	*F*	*gnificance F*
Regression	2	139.308	69.654	14.24983	5.96E-05
Residual	27	131.9775	4.888057		
Total	29	271.2855			

	Coefficients	*andard Err*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	10.7668	3.311688	3.25114	0.003078	3.97174	17.56179	3.97174	17.56179
Income	0.0001	3.7E-05	1.921505	0.065283	-4.8E-06	0.000147	-4.8E-06	0.000147
Years at company	-3.4714	1.070583	-3.24255	0.003145	-5.66808	-1.27477	-5.66808	-1.27477

$$\text{Job Satisfaction} = 10.7668 + 0.0001(\text{Income}) - 3.4714(\text{Years})$$

Multiple Regression with a binary predictor

$$\text{Job Satisfaction} = 10.7668 + 0.0001(\text{Income}) - 3.4714(\text{Years})$$

Person A



Income: 75,000
Years: "Less than 4" (1)

$$\begin{aligned}\text{Job Satisfaction} &= 10.7668 + .0001(75000) - 3.4714(1) \\ &= 10.7668 + 7.5 - 3.4714 \\ &= 14.7953\end{aligned}$$

Person B

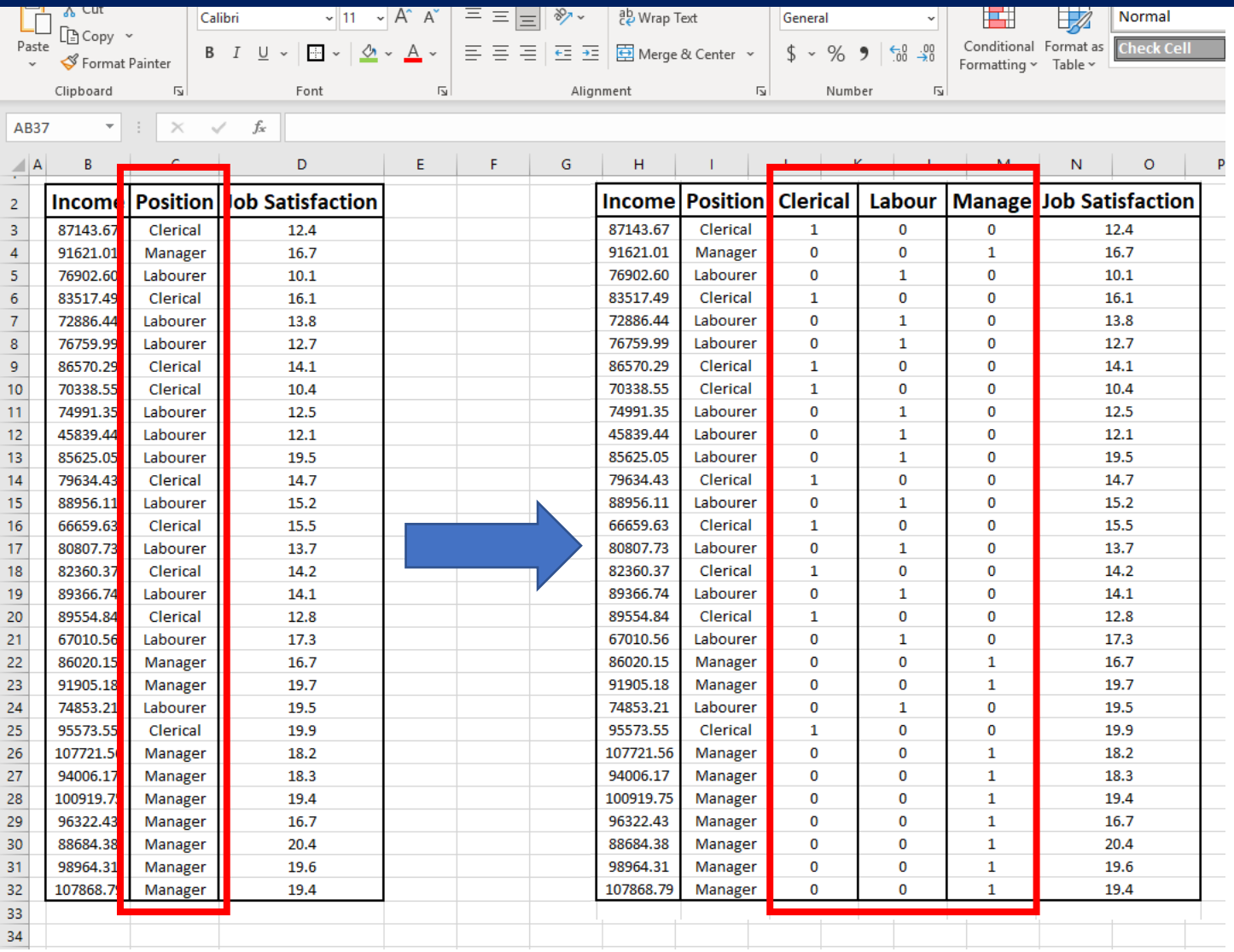


Income: 75,000
Years: "More than 4" (0)

$$\begin{aligned}\text{Job Satisfaction} &= 10.7668 + .0001(75000) - 3.4714(0) \\ &= 10.7668 + 7.5 - 0 \\ &= 18.2667\end{aligned}$$

MONASH University

Multiple Regression with a categorical (k > 2) predictor



Income	Position	Job Satisfaction
87143.67	Clerical	12.4
91621.01	Manager	16.7
76902.60	Labourer	10.1
83517.49	Clerical	16.1
72886.44	Labourer	13.8
76759.99	Labourer	12.7
86570.29	Clerical	14.1
70338.55	Clerical	10.4
74991.35	Labourer	12.5
45839.44	Labourer	12.1
85625.05	Labourer	19.5
79634.43	Clerical	14.7
88956.11	Labourer	15.2
66659.63	Clerical	15.5
80807.73	Labourer	13.7
82360.37	Clerical	14.2
89366.74	Labourer	14.1
89554.84	Clerical	12.8
67010.56	Labourer	17.3
86020.15	Manager	16.7
91905.18	Manager	19.7
74853.21	Labourer	19.5
95573.55	Clerical	19.9
107721.56	Manager	18.2
94006.17	Manager	18.3
100919.75	Manager	19.4
96322.43	Manager	16.7
88684.38	Manager	20.4
98964.31	Manager	19.6
107868.79	Manager	19.4

Income	Position	Clerical	Labour	Manage	Job Satisfaction
87143.67	Clerical	1	0	0	12.4
91621.01	Manager	0	0	1	16.7
76902.60	Labourer	0	1	0	10.1
83517.49	Clerical	1	0	0	16.1
72886.44	Labourer	0	1	0	13.8
76759.99	Labourer	0	1	0	12.7
86570.29	Clerical	1	0	0	14.1
70338.55	Clerical	1	0	0	10.4
74991.35	Labourer	0	1	0	12.5
45839.44	Labourer	0	1	0	12.1
85625.05	Labourer	0	1	0	19.5
79634.43	Clerical	1	0	0	14.7
88956.11	Labourer	0	1	0	15.2
66659.63	Clerical	1	0	0	15.5
80807.73	Labourer	0	1	0	13.7
82360.37	Clerical	1	0	0	14.2
89366.74	Labourer	0	1	0	14.1
89554.84	Clerical	1	0	0	12.8
67010.56	Labourer	0	1	0	17.3
86020.15	Manager	0	0	1	16.7
91905.18	Manager	0	0	1	19.7
74853.21	Labourer	0	1	0	19.5
95573.55	Clerical	1	0	0	19.9
107721.56	Manager	0	0	1	18.2
94006.17	Manager	0	0	1	18.3
100919.75	Manager	0	0	1	19.4
96322.43	Manager	0	0	1	16.7
88684.38	Manager	0	0	1	20.4
98964.31	Manager	0	0	1	19.6
107868.79	Manager	0	0	1	19.4

- We need to choose one of the levels to be a reference category, e.g.:

$$b_0 + b_1\text{Income} + b_2\text{Clerical} + b_3\text{Labourer}$$

- Here we have chosen “Manager” to be the reference category, so it does not have a term in the model.

MONASH University

Multiple Regression with a categorical (k > 2) predictor

$$X = \begin{bmatrix} 1 & \text{Income}_A & 1 & 0 & 0 \\ 1 & \text{Income}_B & 0 & 1 & 0 \\ 1 & \text{Income}_C & 0 & 0 & 1 \end{bmatrix}$$

- For every row, exactly one of the three dummies equals 1 and the other two equal 0

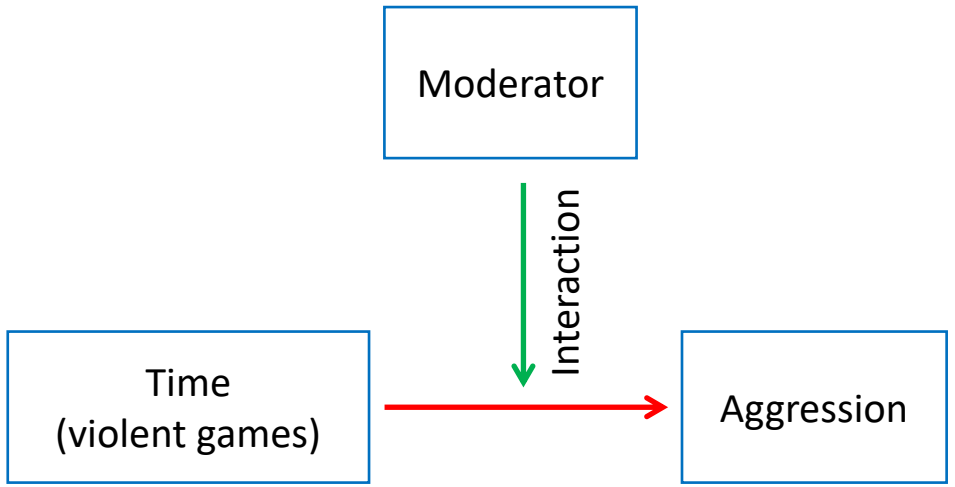
  Clerical + Manager + Labourer = 1 = Intercept

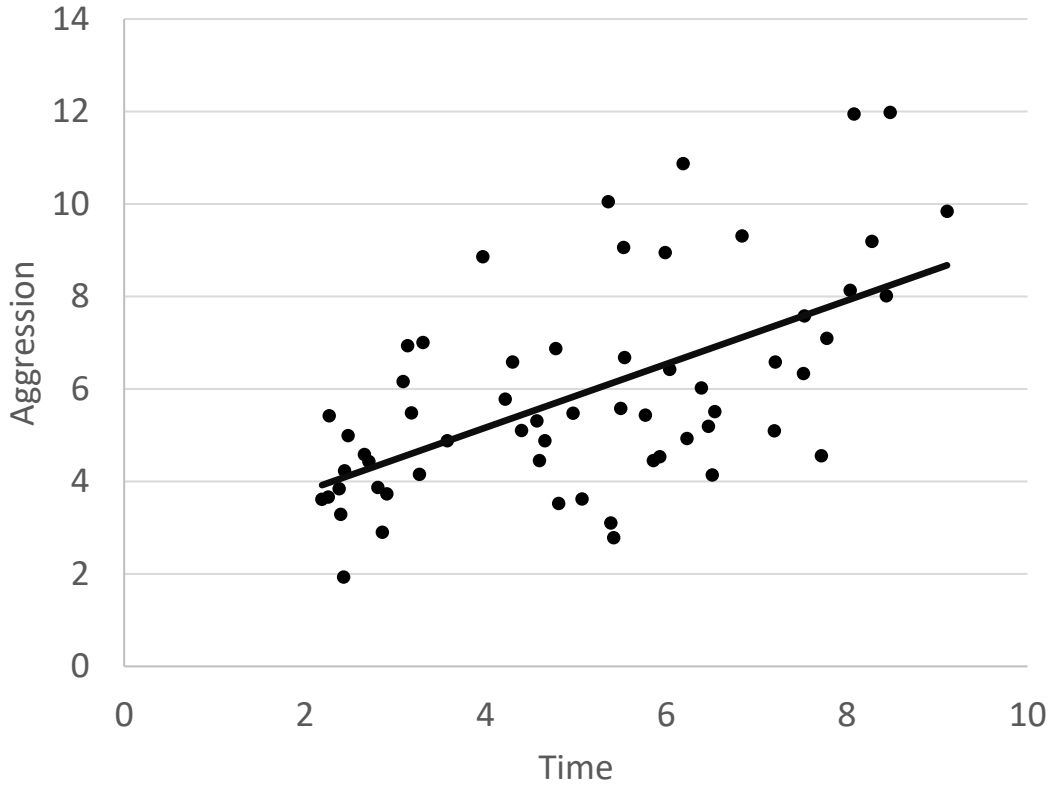
- That means column 3 + column 4 + column 5 = column 1
  (i.e. one column is an exact linear combination of others, the matrix is **rank-deficient**)

- $X^T X$ is singular, so OLS regression cannot compute unique coefficient estimates.

Interaction Effects

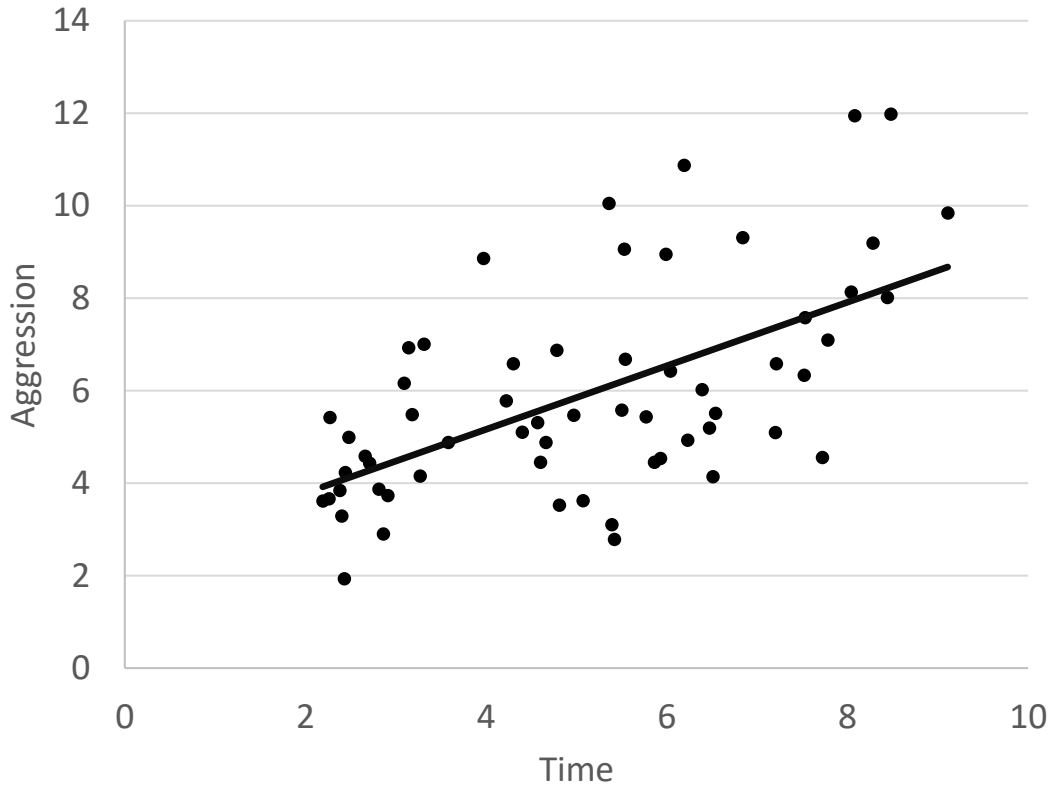
INTERACTION

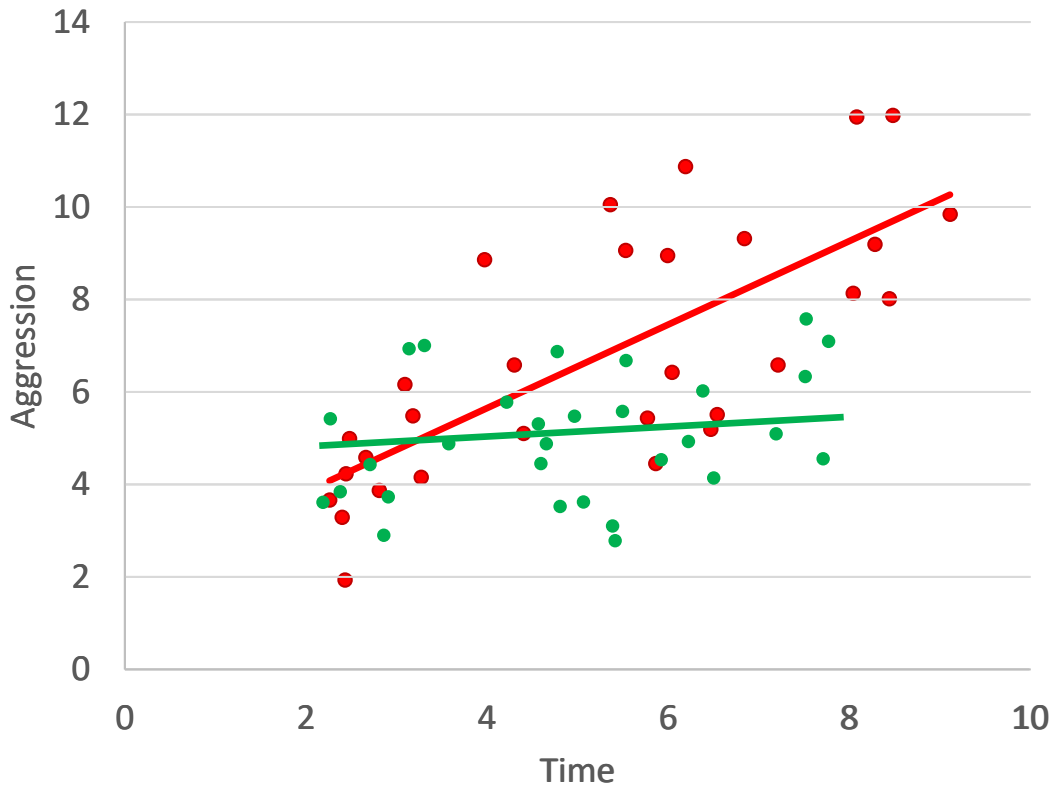




MONASH University

Interaction Effects



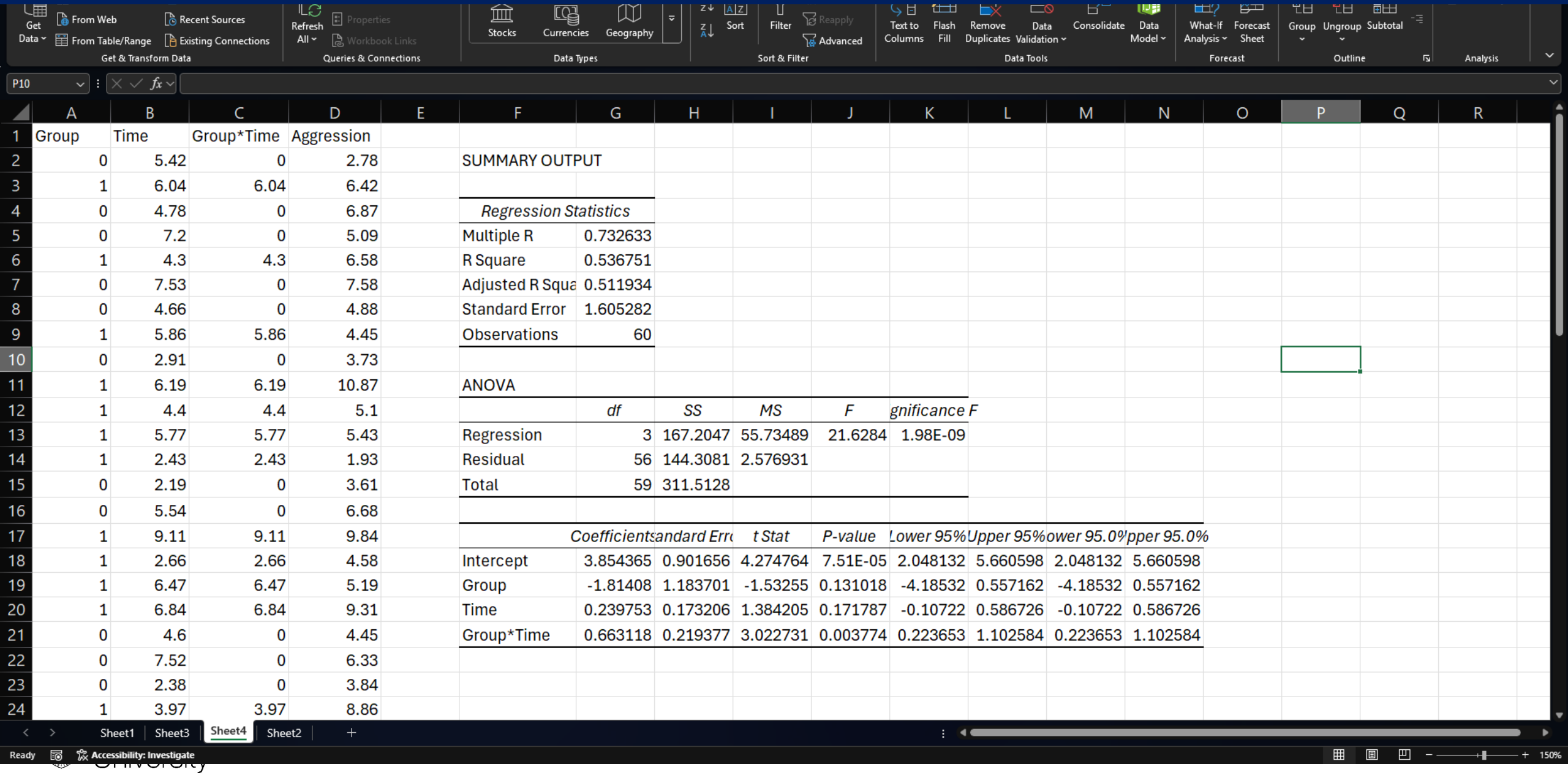


MONASH University

Interaction Effects

- The combined effect of two variables on another is conceptually known as moderation (or interaction)
- Mathematically, a moderator is the product of two variables (e.g. *variable 1* x *variable 2*)
- We can use Excel to determine whether or not a moderator is significant using the Data Analysis tool kit
- However, to obtain individual regression equations we'll have to compute a moderator and do some (simple) maths

Interaction Effects



Group	Time	Group*Time	Aggression
0	5.42	0	2.78
1	6.04	6.04	6.42
0	4.78	0	6.87
0	7.2	0	5.09
1	4.3	4.3	6.58
0	7.53	0	7.58
0	4.66	0	4.88
1	5.86	5.86	4.45
0	2.91	0	3.73
1	6.19	6.19	10.87
1	4.4	4.4	5.1
1	5.77	5.77	5.43
1	2.43	2.43	1.93
0	2.19	0	3.61
0	5.54	0	6.68
1	9.11	9.11	9.84
1	2.66	2.66	4.58
1	6.47	6.47	5.19
1	6.84	6.84	9.31
0	4.6	0	4.45
0	7.52	0	6.33
0	2.38	0	3.84
1	3.97	3.97	8.86

SUMMARY OUTPUT

Regression Statistics

Multiple R	0.732633
R Square	0.536751
Adjusted R Squa	0.511934
Standard Error	1.605282
Observations	60

ANOVA

	df	*SS*	*MS*	*F*	*gnificance F*
Regression	3	167.2047	55.73489	21.6284	1.98E-09
Residual	56	144.3081	2.576931		
Total	59	311.5128			

	Coefficients	*tandard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	3.854365	0.901656	4.274764	7.51E-05	2.048132	5.660598	2.048132	5.660598
Group	-1.81408	1.183701	-1.53255	0.131018	-4.18532	0.557162	-4.18532	0.557162
Time	0.239753	0.173206	1.384205	0.171787	-0.10722	0.586726	-0.10722	0.586726
Group*Time	0.663118	0.219377	3.022731	0.003774	0.223653	1.102584	0.223653	1.102584

Interaction Effects

	Coefficients	Standard Err	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	3.854365	0.901656	4.274764	7.51E-05	2.048132	5.660598	2.048132	5.660598
Group	-1.81408	1.183701	-1.53255	0.131018	-4.18532	0.557162	-4.18532	0.557162
Time	0.239753	0.173206	1.384205	0.171787	-0.10722	0.586726	-0.10722	0.586726
Group*Time	0.663118	0.219377	3.022731	0.003774	0.223653	1.102584	0.223653	1.102584

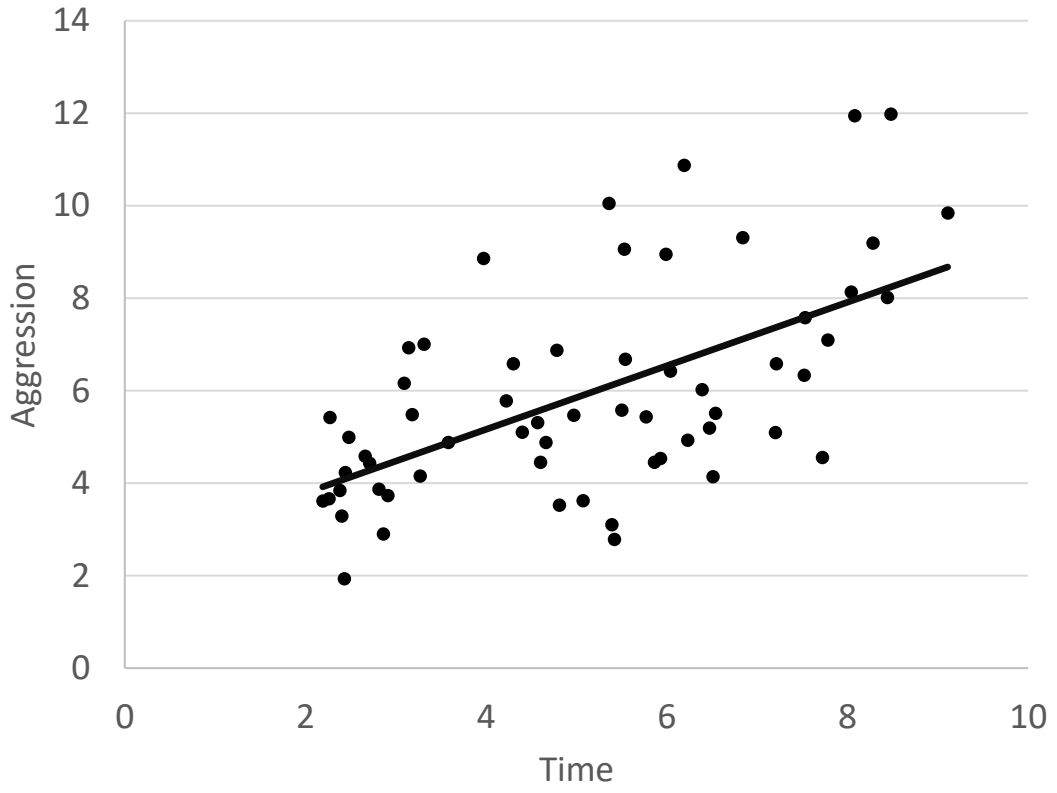
$$\text{Aggression} = 3.85 - 1.81(\text{Group}) + 0.24(\text{Time}) + 0.66(\text{Group} \times \text{Time})$$

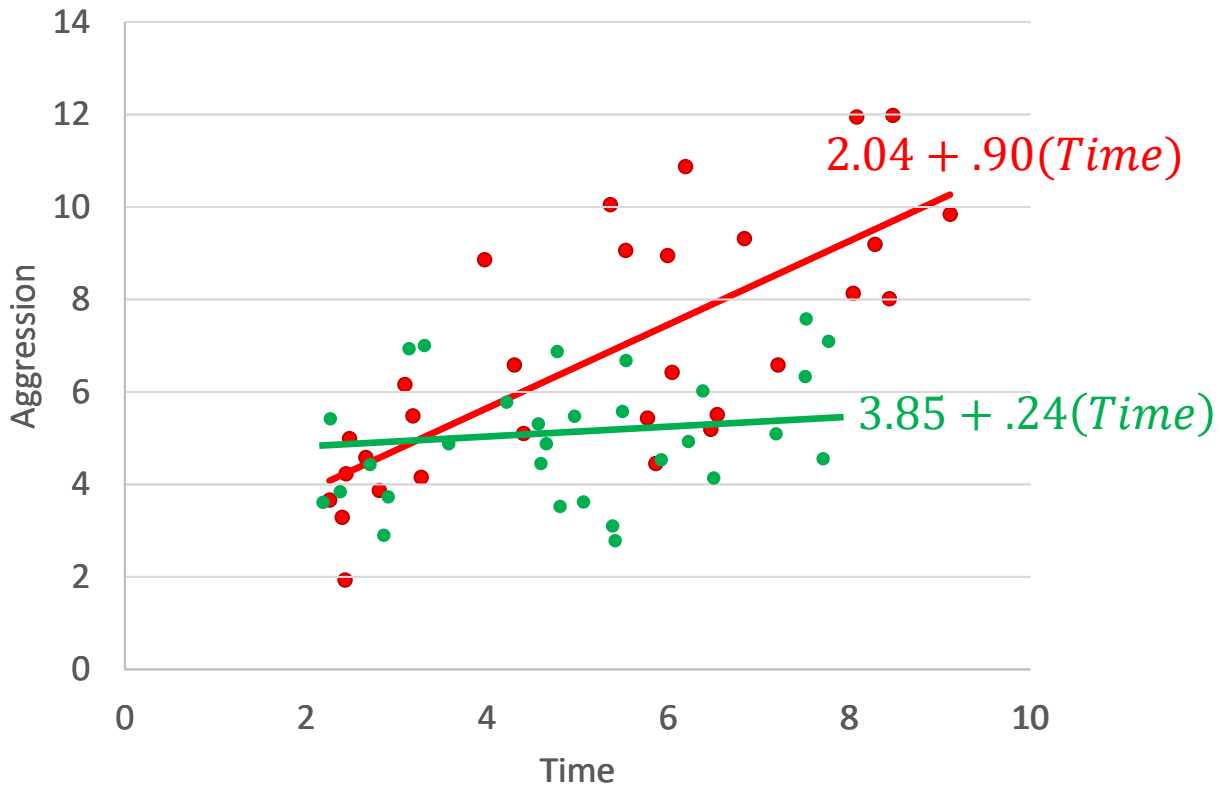
$$\begin{aligned} Aggression_{Group(0)} &= 3.85 - 1.81(Group) + .24(Time) + .66(Group \times Time) \\ &= 3.85 - 1.81(0) + .24(Time) + .66(0 \times Time) \\ &= 3.85 - 1.81(0) + .24(Time) + .66(0 \times Time) \\ &= 3.85 + .24(Time) \end{aligned}$$

$$\begin{aligned} Aggression_{Group(1)} &= 3.85 - 1.81(Group) + .24(Time) + .66(Group \times Time) \\ &= 3.85 - 1.81(1) + .24(Time) + .66(1 \times Time) \\ &= 3.85 - 1.81 + .24(Time) + .66(Time) \\ &= 2.04 + .90(Time) \end{aligned}$$

MONASH University

Interaction Effects



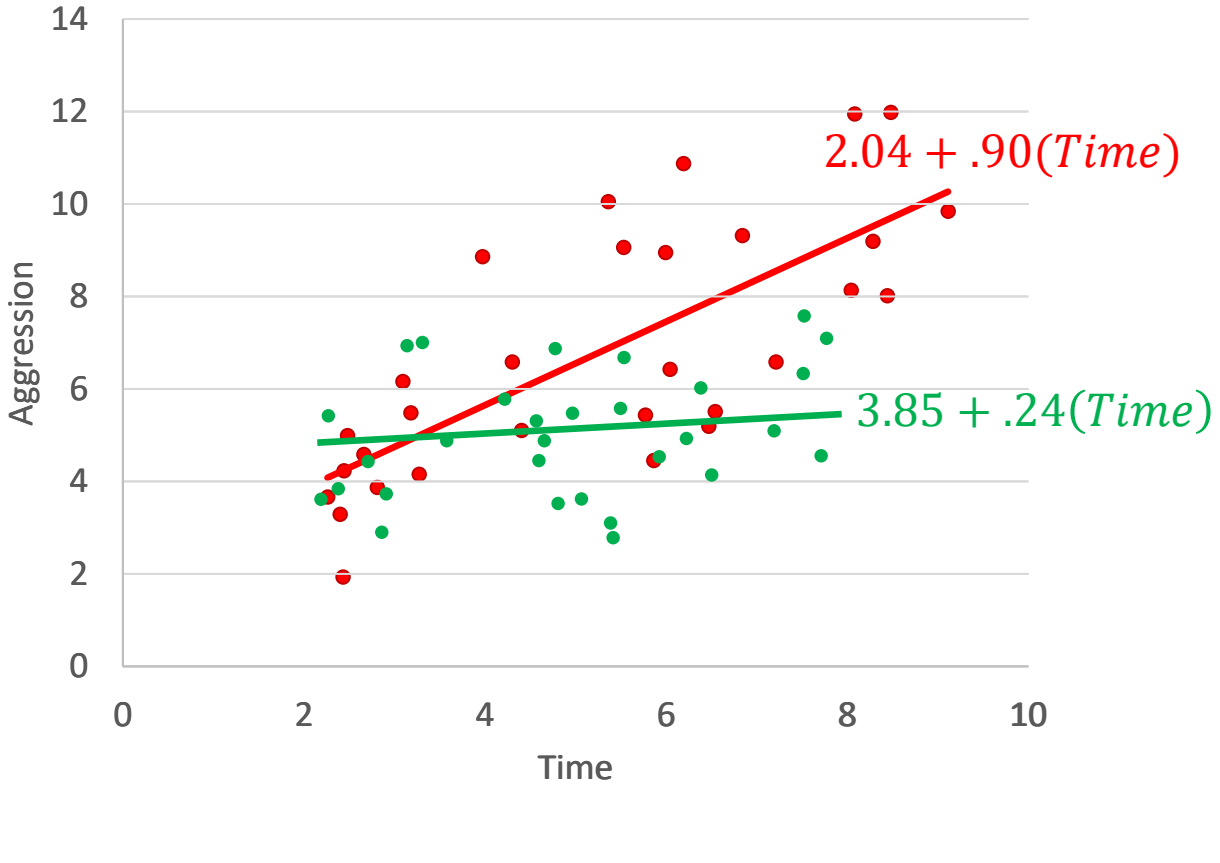


MONASH University

Interaction Effects

	Coefficients	tandard Erro	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	3.854365	0.901656	4.274764	7.51E-05	2.048132	5.660598	2.048132	5.660598
Group	-1.81408	1.183701	-1.53255	0.131018	-4.18532	0.557162	-4.18532	0.557162
Time	0.239753	0.173206	1.384205	0.171787	-0.10722	0.586726	-0.10722	0.586726
Group*Time	0.663118	0.219377	3.022731	0.003774	0.223653	1.102584	0.223653	1.102584

$$\text{Aggression} = 3.85 - 1.81(\text{Group}) + 0.24(\text{Time}) + 0.66(\text{Group} \times \text{Time})$$



MONASH University